5 Design of Digital Filters

Digital Filter

- It is a mathematical algorithm implemented into hardware/software, which operates on digital input and produces a digital output.
- The objective is to extract information from signals or to improve the quality of a signal.

Applications of Digital Filters

- Data compression
- Speech processing
- Image processing etc.

Advantages of Digital Filters

1. The performance of digital filters will not be altered with any change in environment, e.g., thermal variation.
2. These posses linear phase characteristics.
3. Response of a digital filter can be adjusted by using a programmable processor.
4. Digital filters can be used at low frequency.
5. Various input signals can be filtered by using a single digital filter without making any changes in the hardware.

Disadvantages of Digital Filters

- There is speed limitation.
- There is finite word length effect.
- It has a limited bandwidth.

Classification of Digital Filters

- Finite Impulse Response (FIR)
- Infinite Impulse Response (IIR)
- **FIR:** The impulse response sequence is of finite duration, i.e., it has finite number of non-zero terms.
  For example, $h(n) = 2, |n| \leq 4$
  $= 0$, elsewhere

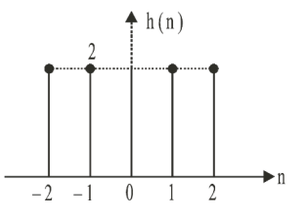


Fig. 5.1

- **IIR:** The impulse response sequence is of infinite duration, i.e., infinite number of non-zero terms.
  For example, $h(n) = a^n u(n), n \geq 0$

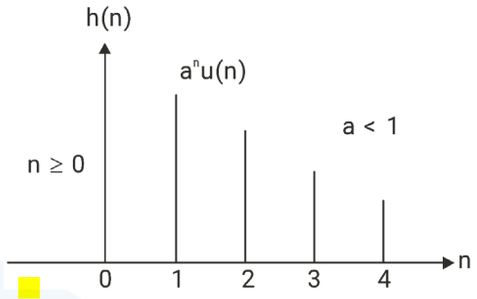


Fig. 5.2

- Digital filter can be expressed into difference equation form.

i.e., $y(n) = -\sum_{k=1}^{N} a_k y(n-k) + \sum_{k=0}^{M} b_k \cdot x(n-k)$

An FIR filter length M can be expressed into difference equation form by:

$$y(n) = \sum_{k=0}^{M-1} b_k \cdot x(n-k)$$

Advantages of FIR over IIR

- It possesses linear phase characteristics.
- It is always stable.
- Linear phase characteristics means phase delay and group delay must be constant, i.e., independent of frequency.
  Then, $h(n) = h(M-1-n)$ symmetry, where M = length of filter
  or
  $h(n) = -h(M-1-n)$ Anti-symmetry

Magnitude and Phase Response of Digital Filters

We know that response of an FIR filter having length M is defined as:

$$y(n) = \sum_{k=0}^{M-1} b_k \cdot x(n-k)$$

or, $h(n) = b_n, \ldots 0 < n \leq M-1$

$$H(e^{j\omega}) = \sum_{n=0}^{M-1} h(n) \cdot e^{-j\omega n}$$

Magnitude of $H(e^{j\omega})$ is given by:

$$\left|H(e^{j\omega})\right| = \sqrt{Re\left[H\left(e^{j\omega}\right)\right]^2 + Im\left[H\left(e^{j\omega}\right)\right]^2}$$

and phase response,

$$\phi(\omega) = \tan^{-1}\left[\frac{Im\left[H\left(e^{j\omega}\right)\right]}{Re\left(H(e^{j\omega})\right)}\right]$$

The phase and group delay is defined as:

$$t_p = \frac{-\phi(\omega)}{\omega} \;\&\; t_g = \frac{-d\phi(\omega)}{d\omega}$$

Filter will have linear phase only when phase delay and group delay are constant.

i.e., $$-\frac{\phi(\omega)}{\omega} = \tau\,, -\pi \le \omega \le \pi$$

Where τ: constant phase delay expressed in the number of samples.

or, $$\phi(\omega) = -\omega\tau$$

$$\tan^{-1}\left[\frac{Im\left[H(e^{j\omega})\right]}{R_e\left[H(e^{j\omega})\right]}\right] = -\omega\tau$$

or

$$\tan^{-1}\left[\frac{\sum_{n=0}^{M-1} h(n)\sin\omega n}{\sum_{n=0}^{M-1} h(n)\cos\omega n}\right] = \omega\tau$$

or, $$\left[\frac{\sum_{n=0}^{M-1} h(n)\cdot\sin\omega n}{\sum_{n=0}^{M-1} h(n)\cdot\cos\omega n}\right] = \tan\omega\tau = \frac{\sin\omega\tau}{\cos\omega\tau}$$

$$\sum_{n=0}^{M-1} h(n)\cdot\sin\omega n\cdot\cos\omega\tau = \sum_{n=0}^{M-1} h(n)\cdot\cos\omega n\cdot\sin\omega\tau$$

$$\sum_{n=0}^{M-1} h(n)\cdot\left[\sin\omega\tau\cdot\cos\omega n - \cos\omega\tau\cdot\sin\omega n\right] = 0$$

$$\sum_{n=0}^{M-1} h(n)\sin\left(\omega\tau - \omega n\right) = 0$$

This is possible only when:

$$\tau = \frac{M-1}{2} \text{ and } h(n) = \pm h(M-1-n)$$

i.e., filter have linear phase only when:

$$\tau = \frac{M-1}{2} \;\&\; h(n) = \pm h\left[M-1-n\right]$$

- **Group delay:** Measure the time delay of amplitude envelope of various sinusoidal components
- **Phase delay:** Measure of the time delay of phase of frequency component of signals are not delayed by the same amount then output of device is a distorted signal.

  Group delay $\tau_g = \frac{-d\phi(\omega)}{d\omega} \Rightarrow$ Measure of phase response at any given time.
- If the system is linear, then group and phase delay are constant and equal to overall delay of the system.

Magnitude Characteristics and Order of FIR Filters

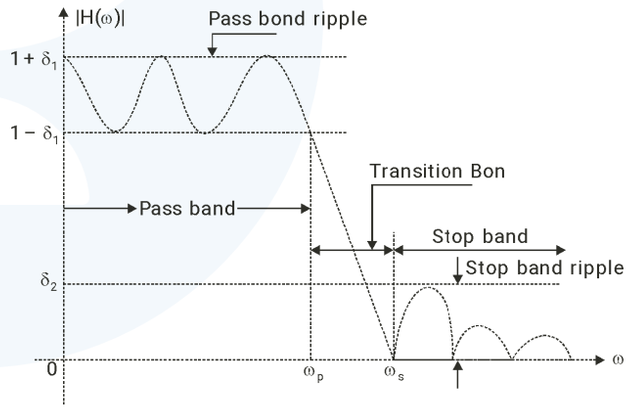


Fig. 5.3

δ_1 = Pass band ripple

δ_2 = Stop band ripple

ω_p = Pass band edge frequency

ω_s = Stop band edge frequency

Magnitude of Filter response varies between $1+\delta_1$ to $1-\delta_1$

The order of filters is given by:

$$N = \frac{-10\log_{10}\left(\delta_1\cdot\delta_2\right) - 15}{14\,\Delta F}$$

Difference between FIR and IIR Filters

The difference between FIR and IIR filters are as follows.

1. Poles of FIR filters are all located at the origin. Therefore, the FIR filters are always stable. The stability of IIR filters depends on the location of the poles. The IIR filters are stable, if all its poles lie inside the unit circle.
2. FIR filters tend to be less sensitive to finite word length effects. The quantised IIR filters can become unstable.
3. The FIR filters can have an exactly linear phase response. A linear phase filter does not distort a signal within the pass band. It only delays it. The phase response of IIR filters are non-linear, especially at the band edges.
4. FIR filters are realised non-recursively, whereas the IIR filters are realised recursively.

Note:

- FIR filters can be used if no phase distortion is desired and the number of filter coefficients is not too large.
- If some phase distortion is either tolerable or unimportant, an IIR filter is preferable primarily because its implementation involves fewer parameters, which require less memory and have lower computation.

Conditions for Filters to Have a Linear Phase

Transfer function of FIR filter of length M is given as:

$$H(z) = \sum_{n=0}^{M-1} h(n) z^n$$

$$H(e^{j\omega}) = \sum_{n=0}^{M-1} h(n) e^{-j\omega n} = \sum_{n=0}^{M-1} [h(n)\cos\omega n - j h(n)\sin\omega n]$$

$$H(e^{j\omega}) = \left|H(e^{j\omega})\right| e^{j\underline{|H(e^{j\omega})}}$$

$$M = \left|H(e^{j\omega})\right|$$

$$\theta(\omega) = \underline{|H(e^{j\omega})} = \tan^{-1}\left[\frac{-\sum_{n=0}^{M-1} h(n)\sin\omega n}{\sum_{n=0}^{M-1} h(n)\cos\omega n}\right]$$

The phase and group delay of the filter are:

$$t_p = -\frac{\theta(\omega)}{\omega},\ t_g = -\frac{d\theta(\omega)}{d\omega}$$

For phase response to be linear:

$$\theta(\omega) = -\tau\omega \qquad -\pi < \omega < \pi$$

$$\boxed{t_p = t_g = \tau}$$

$$\theta(\omega) = -\omega\tau = \tan^{-1}\left[\frac{-\sum_{n=0}^{M-1} h(n)\sin\omega n}{\sum_{n=0}^{M-1} h(n)\cos\omega n}\right]$$

$$\tan\omega\tau = \frac{\sin\omega\tau}{\cos\omega\tau} = \frac{\sum_{n=0}^{M-1} h(n)\sin\omega n}{\sum_{n=0}^{M-1} h(n)\cos\omega n}$$

$$\sum_{n=0}^{M-1} h(n)\cos\omega n\sin\omega\tau - \sum_{n=0}^{M-1} h(n)\sin\omega n\cos\omega\tau = 0$$

$$\sum_{n=0}^{M-1} h(n)\sin(\omega\tau - \omega n) = 0$$

The solution of this equation is

$$\tau = \frac{M-1}{2} \qquad \text{... (i)}$$

$$h(n) = h(M-1-n),\ 0 \le n \le M-1 \qquad \text{... (ii)}$$

Note: FIR filters have constant phase and groups delay, if condition (i) and (ii) are satisfied. Impulse response h(n) is symmetric about τ, which is the centre of symmetry.

Case I: M-odd

$\tau = \frac{M-1}{2}$ is an integer and impulse response is symmetrical about the sample point $\frac{M-1}{2}$

For example, M = 7

$h(n) = h(6-n),\ 0 \le n \le 6$

h(0) = h(6), h(1)= h(5), h(2) = h(4) and h(3) = h(3)

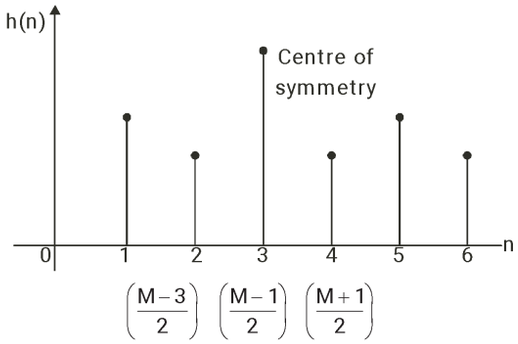


Fig. 5.4

Case II: M-even

$\tau = \frac{M-1}{2}$ is not an integer and the impulse response is symmetrical about the point that lies in the centre between coefficients $\frac{M-1}{2}$ and $\frac{M}{2}$.

Ex. M = 8

$h(n) = h(7 - n),\ 0 \le n \le 7$

h(0) = h(7), h(1) = h(6), h(2) = h(5) and h(3) = h(4)

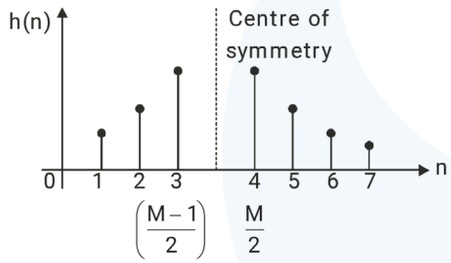


Fig. 5.5

In application, where only constant group delay is desired, the phase response can have the form as:

$\theta(\omega) = \beta - \omega\tau$

$$t_g = -\frac{d\theta(\omega)}{d\omega} = \tau\ (\text{constant})$$

$$\theta(\omega) = \beta - \omega\tau = \tan^{-1}\left[\frac{-\sum_{n=0}^{M-1} h(n)\sin\omega n}{\sum_{n=0}^{M-1} h(n)\cos\omega n}\right]$$

$$\tan(\beta - \omega\tau) = \frac{\sin(\beta - \omega\tau)}{\cos(\beta - \omega\tau)} = \frac{-\sum_{n=0}^{M-1} h(n)\sin\omega n}{\sum_{n=0}^{M-1} h(n)\cos\omega n}$$

$$\sum_{n=0}^{M-1} h(n)\cos\omega n \sin(\beta - \omega\tau) + \sum_{n=0}^{M-1} h(n)\sin\omega n \cos(\beta - \omega\tau) = 0$$

$$\sum_{n=0}^{M-1} h(n)\sin[\beta - (\tau - n)\omega] = 0$$

if $\beta = \frac{\pi}{2}$

$$\boxed{\sum_{n=0}^{M-1} h(n)\cos(\tau - n)\omega = 0}$$

The solution of the above equation:

$$\boxed{\tau = \frac{M-1}{2}}$$

$$\boxed{h(n) = -h(M-1-n),\ 0 \le n \le M-1}$$

Case I: M-odd

$\tau = \frac{M-1}{2}$ is an integer and h(n) is anti-symmetrical about the sample point $\frac{M-1}{2}$ and $h\left(\frac{M-1}{2}\right) = 0$ {sample at $\tau = \frac{M}{2} - 1$ must be equal to zero}

For example, M = 7

$h(n) = -h(6 - n)$

h(0) = −h(6), h(1) = −h(5), h(2) = −h(4), h(3) = − h(3) = 0

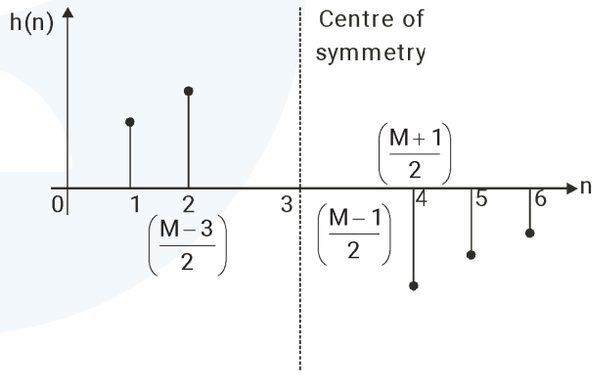


Fig. 5.6

Case II: M-even

$\tau = \frac{M-1}{2}$ is not an integer and h(n) is anti-symmetrical about the point lies in the centre between the coefficients $\frac{M}{2} - 1$ and $\frac{M}{2}$

For example, M = 8

$h(n) = -h(7 - n),\ 0 \le n \le 7$

h(0) = −h(7), h(1) = −h(6), h(2) = −h(5), h(3) = −h(t)

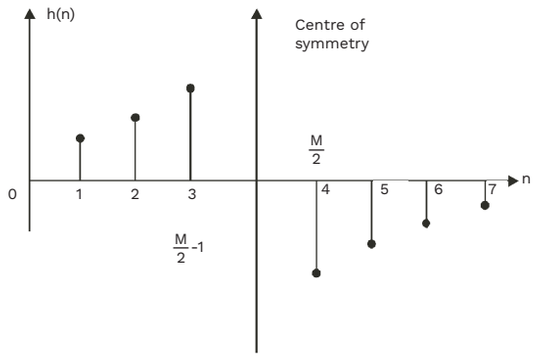


Fig. 5.7

Frequency Response of Linear Phase FIR Filter

$$H(e^{j\omega}) = H_r(e^{j\omega})\, e^{j\theta(\omega)}$$

$$\theta(\omega) = \beta - \omega\tau$$

$= 0$ for symmetric response

$= \frac{\pi}{2}$ for anti-symmetric response

- $H_r(e^{j\omega})$ → Amplitude response or zero phase response and not the magnitude response.
- Amplitude response is a real function and it may be both positive and negative.
- The magnitude response is always positive.

1. Type 1: Symmetric impulse response with odd length (M odd)

$\beta = 0,\ \tau = \frac{M-1}{2}$ is an integer.

$h(n) = h(M-1-n),\ 0 \le n \le M-1$

$$H(e^{j\omega}) = \sum_{n=0}^{M-1} h(n)e^{-j\omega n}$$

$$= \sum_{n=0}^{\frac{M-3}{2}} h(n)e^{-j\omega n} + h\left(\frac{M-1}{2}\right)e^{-j\omega\left(\frac{M-1}{2}\right)} + \sum_{n=\left(\frac{M+1}{2}\right)}^{M-1} h(n)e^{-j\omega n}$$

$$= \sum_{n=0}^{\frac{M-3}{2}} h(n)e^{-j\omega n} + h\left(\frac{M-1}{2}\right)e^{-j\omega\left(\frac{M-1}{2}\right)}.$$

$$+ \sum_{n=0}^{M-1} \underbrace{h(M-1-n)}_{h(n)} e^{-j\omega(M-1-n)}$$

$$H\left(e^{j\omega}\right) = \sum_{n=0}^{\frac{M-3}{2}} h(n)\left[e^{-j\omega n} + e^{-j\omega(M-1-n)}\right] + h\left(\frac{M-1}{2}\right)e^{-j\omega\frac{(M-1)}{2}}$$

$$= e^{-j\omega\left(\frac{M-1}{2}\right)}\left\{\sum_{n=0}^{\frac{M-3}{1}} h(n)\left[e^{-j\omega\left(\frac{M-1}{2}-n\right)} + e^{-j\omega\left(\frac{M-1}{2}-n\right)}\right] + h\left(\frac{M-1}{2}\right)\right\}$$

$$= e^{-j\omega\left(\frac{M-1}{2}\right)}\left\{\sum_{n=0}^{\frac{M-3}{2}} 2h(n)\cos\left[\left(\frac{M-1}{2}-n\right)\omega\right] + h\left(\frac{M-1}{2}\right)\right\}$$

Let, $k = \frac{M-1}{2} - n$

$$H\left(e^{j\omega}\right) = e^{-j\omega\left(\frac{M-1}{2}\right)}\left\{\sum_{k=1}^{\frac{(M-1)}{2}} 2h\left(\frac{M-1}{2}-k\right)\cos(k\omega) + h\left(\frac{M-1}{2}\right)\right\}$$

$$H\left(e^{j\omega}\right) = e^{-j\omega\left(\frac{M-1}{2}\right)}\sum_{k=0}^{\left(\frac{M-1}{2}\right)} a(k)\cos(k\omega) = e^{j\theta(\omega)}H_r\left(e^{j\omega}\right)$$

Where, $a(0) = h\left(\frac{M-1}{2}\right)$

$$a(k) = 2h\left(\frac{M-1}{2}-k\right),\ 1 \le k \le \frac{M-1}{2}$$

$$\theta(\omega) = -\omega\left(\frac{M-1}{2}\right)$$

$$\tau = \frac{-\theta(\omega)}{\omega} = \frac{M-1}{2}$$

$$\boxed{H_r\left(e^{j\omega}\right) = \sum_{k=0}^{\frac{M-1}{2}} a(k)\cos(k\omega)}$$ $H_r(e^{j0})$ & $H_r(e^{j\pi})$ are non-zero

- This type is most widely used for filter design because it allows to design any filter type by appropriate choice of filter coefficients.

2. Type 2: Symmetric impulse response with even length (M even)

$\beta = 0,\ \tau = \dfrac{M-1}{2}$ is not an integer length.

$$H\left(e^{j\omega}\right) = \sum_{n=0}^{M-1} h(n)e^{-j\omega n}$$

$$= \sum_{n=0}^{\frac{M}{2}-1} h(n)e^{-j\omega n} + \sum_{n=\frac{M}{2}}^{M-1} h(n)e^{-j\omega n}$$

$$= \sum_{n=0}^{\frac{M}{2}-1} h(n)e^{-j\omega n} + \sum_{n=0}^{\frac{M}{2}-1} \underbrace{h(M-1-n)}_{h(n)}\, e^{-j\omega(M-1-n)}$$

$$= \sum_{n=0}^{\frac{M}{2}-1} h(n)\left[e^{-j\omega n} + e^{-j\omega(M-1-n)}\right]$$

$$= e^{-j\omega\left(\frac{M-1}{2}\right)} \sum_{n=0}^{\frac{M}{2}-1} h(n)\left[e^{j\omega\left(\frac{M-1}{2}-n\right)} + e^{-j\omega\left(\frac{M-1}{2}-n\right)}\right]$$

$$= e^{-j\omega\left(\frac{M-1}{2}\right)} \sum_{n=0}^{\frac{M}{2}-1} 2h(n)\cos\left[\omega\left(\frac{M-1}{2}-n\right)\right]$$

Let, $k = \dfrac{M}{2} - n$

$$H\left(e^{j\omega}\right) = e^{-j\omega\left(\frac{M-1}{2}\right)} \sum_{k=1}^{\frac{M}{2}} 2h\left(\frac{M}{2}-k\right)\cos\left[\omega\left(k-\frac{1}{2}\right)\right]$$

$$= e^{-j\omega\left(\frac{M-1}{2}\right)} \sum_{k=1}^{\frac{M}{2}} b(k)\cos\left[\omega\left(k-\frac{1}{2}\right)\right] = e^{j\theta(\omega)}H_r\left(e^{j\omega n}\right)$$

$$b(k) = 2h\left(\frac{M}{2}-k\right),\ 1 \le k \le \frac{M}{2}$$

$$\tau = -\frac{\theta(\omega)}{\omega} = \frac{M-1}{2}$$

$$H_r\left(e^{j\omega}\right) = \sum_{k=1}^{\frac{M}{2}} b(k)\cos\left[\omega\left(k-\frac{1}{2}\right)\right]$$

$$H_r\left(e^{j\pi}\right) = \sum_{k=1}^{\frac{M}{2}} b(k)\cos\underbrace{\left[\pi\left(k-\frac{1}{2}\right)\right]}_{0} = 0$$

Note: $H_r(e^{j\pi}) = 0$, therefore, we cannot use (i.e., symmetric h(n), M even) to design either a H.P. or B.S.F. (Band Stop Filter)

3. Type 3: Anti-symmetric impulse response with odd length (M odd)

$\beta = \dfrac{\pi}{2},\ \tau = \dfrac{M-1}{2}$ is an integer

$h(n) = -h(M-1-n),\ 0 \le n \le M-1$

$$H\left(e^{j\omega}\right) = \sum_{n=0}^{M-1} h(n)\, e^{-j\omega n}$$

$$= \sum_{n=0}^{\frac{M-3}{2}} h(n)\, e^{-j\omega n} + \underbrace{h\left(\frac{M-1}{2}\right)}_{0} e^{-j\omega\left(\frac{M-1}{2}\right)}$$

$$+ \sum_{n=(M+1)/2}^{M-1} h(n)e^{-j\omega n}$$

$$= \sum_{n=0}^{\frac{M-3}{2}} h(n)\, e^{-j\omega n} + \sum_{n=0}^{\frac{M-3}{2}} \underbrace{h(M-1-n)}_{=-h(n)}\, e^{-j\omega(M-1-n)}$$

$$= \sum_{n=0}^{\frac{M-3}{2}} h(n)\left[e^{-j\omega n} - e^{-j\omega(M-1-n)}\right]$$

$$H\left(e^{j\omega}\right) = je^{-j\omega\left(\frac{M-1}{2}\right)} \sum_{n=0}^{\frac{M-3}{2}} 2h(n)\sin\left[\omega\left(\frac{M-1}{2}\right)-n\right]$$

$$= e^{j\left[\frac{\pi}{2}-\omega\left(\frac{M-1}{2}\right)\right]} \sum_{n=0}^{\frac{M-3}{2}} 2h(n)\sin\left[\omega\left(\frac{M-1}{2}-n\right)\right]$$

Let, $k = \dfrac{M-1}{2} - n$

$$H\left(e^{j\omega}\right) = e^{j\left[\frac{\pi}{2}-\omega\left(\frac{M-1}{2}\right)\right]} \sum_{k=1}^{\frac{M-1}{2}} 2h\left[\left(\frac{M-1}{2}-k\right)\sin(k\omega)\right]$$

$$= e^{j\left[\frac{\pi}{2}-\omega\left(\frac{M-1}{2}\right)\right]} \sum_{k=1}^{\frac{M-1}{2}} c(k)\sin(k\omega)$$

$$= e^{j\theta(\omega)}\, H_r\left(e^{j\omega}\right)$$

$$c(k) = 2h\left(\frac{M-1}{2}-k\right),\ 1 \le k \le \frac{M-1}{2}$$

$$\theta(\omega) = \left[\frac{\pi}{2} - \omega\left(\frac{M-1}{2}\right)\right]$$

$$t_g = \frac{-d\theta(\omega)}{d\omega} = \frac{M-1}{2}$$

$$H_r = \sum_{k=1}^{\frac{M-1}{2}} c(k)\sin(k\omega)$$

$$\boxed{\begin{aligned} H_r(e^{j0}) &= 0 \\ H_r(e^{j\pi}) &= 0 \end{aligned}}$$

- This type [i.e., anti-symmetric h(n), M odd] cannot be used to design either LPF, or HPF, or BSF.

Note: This type is suitable to design Hilbert transform and differentiator.

4. Type 4: Anti-symmetric impulse response with even length (M even)

$$\beta = \frac{\pi}{2}, \tau = \frac{M-1}{2} \text{ is not integer.}$$

$$h(n) = -h(M-1-n),\ 0 \le n \le M-1$$

$$H\left(e^{j\omega}\right) = \sum_{n=0}^{\frac{M}{2}-1} h(n)\, e^{-j\omega n} + \sum_{n=\frac{M}{2}}^{M-1} h(n)\, e^{-j\omega n}$$

$$= \sum_{n=0}^{\frac{M}{2}-1} h(n)\, e^{-j\omega n} + \sum_{n=0}^{\frac{M}{2}-1} h\underbrace{(M-1-n)}_{=-h(n)} e^{-j\omega n(M-1-n)}$$

$$= je^{-j\omega\left(\frac{M}{2}-1\right)} \sum_{n=0}^{\frac{M}{2}-1} 2h(n) \sin\left[\omega\left(\frac{M}{2}-1-n\right)\right]$$

$$= e^{j\left[\frac{\pi}{2}-\omega\left(\frac{M}{2}-1\right)\right]} \sum_{n=0}^{\frac{M}{2}-1} 2h(n) \sin\left[\omega\left(\frac{M}{2}-1-n\right)\right]$$

Let, $k = \frac{M}{2} - n$

$$H\left(e^{j\omega}\right) = e^{j\left[\frac{\pi}{2}-\omega\left(\frac{M-1}{2}\right)\right]} \sum_{k=1}^{\frac{M}{2}} 2h\left(\frac{M}{2}-k\right) \sin\left[\omega\left(\frac{k-1}{2}\right)\right]$$

$$= e^{j\left[\frac{\pi}{2}-\omega\left(\frac{M-1}{2}\right)\right]} \sum_{k=1}^{\frac{M}{2}} d(k) \sin\left[\omega\left(k-\frac{1}{2}\right)\right]$$

$$= e^{j\theta(\omega)} H_r\left(e^{j\omega}\right)$$

$$d(k) = 2h\left(\frac{M}{2}-k\right), 1 \le k \le \frac{M}{2}$$

$$\theta(\omega) = \frac{\pi}{2} - \omega\left(\frac{M-1}{2}\right)$$

$$t_g = \frac{-d\theta(\omega)}{d\omega} = \frac{M-1}{2}$$

$$H_r\left(e^{j\omega}\right) = \sum_{k=1}^{\frac{M}{2}} d(k) \sin\left[\omega\left(k-\frac{1}{2}\right)\right]$$

$$H_r\left(e^{j0}\right) = 0$$

- This type [i.e., anti-symmetric h(n), M even] cannot be used to design either LPF or BPF.

Amplitude Response Characteristics of Four Types of Linear Phase Response

Type	h(n)	M	$H_r\left(e^{j\omega}\right)$	$H_r\left(e^{j\omega}\right)$	Application
1	Symmetric	Odd	$\sum_{k=0}^{\frac{M-1}{2}} a(k)\cos(k\omega)$	Symmetric about $\omega = 0$ & π	All type of filter only sequence for BSF
2	Symmetric	Even	$\sum_{k=1}^{\frac{M}{2}} b(k)\cos\left[\omega\left(k-\frac{1}{2}\right)\right]$	Symmetric about $\omega = 0$ & antisymmetric about $\omega = \pi$ $H_r\left(e^{j\pi}\right) = 0$	Only LPF & BPF
3	Antisymmetric	Odd	$\sum_{k=1}^{\frac{M-1}{2}} c(k)`\sin(k\omega)$	Antisymmetric about both $\omega = 0$ & π $H_r(e^{j0}) = 0$ $H_r(e^{j\pi}) = 0$	BPF, differentiator & Hilbert transformer
4	Antisymmetric	Even	$\sum_{k=1}^{\frac{M}{2}} d(k)\sin\left[\omega\left(k-\frac{1}{2}\right)\right]$	Antisymmetric about $\omega = 0$ and symmetric about $\omega = \pi$ $H_r\left(e^{j0}\right) = 0$	HPF, BPF, differentiator &Hilbert transformer

Table 5.1

Solved Examples

1. Does $H(z) = 1 + 2z^{-1} + 2z^{-2} + z^{-3}$ describe a linear phase filter? Determine its frequency response and calculate the phase delay and group delay.

Solution:

$H(z) = 1 + 2z^{-1} + 2z^{-2} + z^{-3}$

$h(n) = \delta(n) + 2\delta(n-1) + 2\delta(n-2) + \delta(n-3)$

$= \{1, 2, 2, 1\}$

$\uparrow$

$M = 4$

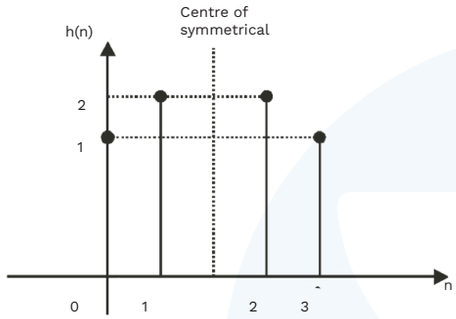


Fig. 5.8

$$H(z)\big|_{z=e^{j\omega}} = 1 + 2e^{-j\omega} + 2e^{-j2\omega} + e^{-j3\omega}$$

$$= e^{\frac{-j3\omega}{2}}\left[e^{\frac{j3\omega}{2}} + 2e^{\frac{j\omega}{2}} + e^{\frac{j\omega}{2}} + e^{\frac{-j3\omega}{2}}\right]$$

$$= e^{\frac{-j3\omega}{2}}\left[4\cos\left(\frac{\omega}{2}\right) + 2\cos\left(\frac{3\omega}{2}\right)\right]$$

$$= e^{j\theta(\omega)}H_r(\omega)$$

$$\theta(\omega) = \frac{-3\omega}{2}$$

$$t_p = \frac{-\theta(\omega)}{\omega} = \frac{3}{2}$$

$$t_g = \frac{-d\theta(\omega)}{d\omega} = \frac{-3}{2}$$

Impulse Response of Ideal Filters

Ideal Low Pass Filter

$$H_{LP}\left(e^{j\omega}\right) = \begin{cases} 1 & , \ |\omega| \le \omega_c \\ 0 & , \ \omega_c < |\omega| \le \pi \end{cases}$$

$$h_{LP}(n) = \frac{\sin \omega_c n}{n\pi} \quad , \ -\infty < n < \infty$$

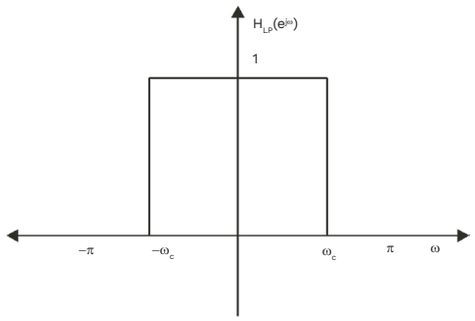


Fig. 5.9

Ideal High Pass Filter

$$H_{HP}\left(e^{j\omega}\right) = \begin{cases} 0 & , \ |\omega| < \omega_c \\ 1 & , \ \omega_c \le |\omega| \le \pi \end{cases}$$

$$h_{HP}(n) = \begin{cases} 1 - \dfrac{\omega_c}{\pi} & \text{for} \quad n = 0 \\ -\dfrac{\sin(\omega_c n)}{n\pi} & \text{for} \quad n > 0 \end{cases}$$

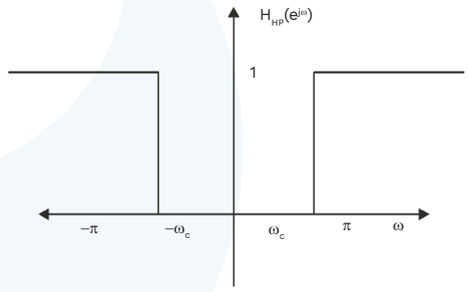


Fig. 5.10

Ideal Band Pass Filter

$$H_{BP}\left(e^{j\omega}\right) = \begin{cases} 0 & , \ |\omega| < \omega_{c_1} \\ 1 & , \ \omega_{c_1} \le |\omega| \le \omega_{c_2} \\ 0 & , \ \omega_{c_2} < |\omega| \le \pi \end{cases}$$

$$h_{BP}(n) = \frac{\sin\left(\omega_{c_2} n\right)}{n\pi} - \frac{\sin\left(\omega_{c_1} n\right)}{n\pi}, \ (n) \ge 0$$

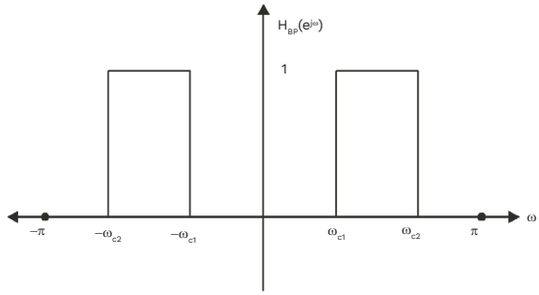


Fig. 5.11

Ideal Band Stop Filter

$$H_{BS}\left(e^{j\omega}\right) = \begin{cases} 1 & |\omega| \le \omega_{c_1} \\ 0 & \omega_{c_1} < |\omega| < \omega_{c_2} \\ 1 & \omega_{c_2} < |\omega| \le \pi \end{cases}$$

$$h_{BS}(n) = \begin{cases} 1 - \dfrac{\left(\omega_{c_2} - \omega_{c_1}\right)}{\pi} & \text{for } n = 0 \\ \dfrac{\sin\left(\omega_{c_1} n\right)}{n\pi} - \dfrac{\sin\left(\omega_{c_2} n\right)}{n\pi} & \text{for } |n| > 0 \end{cases}$$

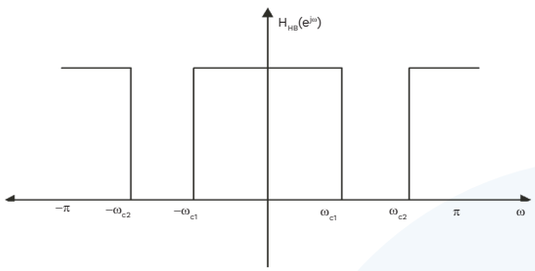


Fig. 5.12

Ideal Hilbert Transformer

The ideal Hilbert transformer is 90° phase shifter.

$$H_{HT}(e^{j\omega}) = \begin{cases} j, & -\pi < \omega < 0 \\ -j, & 0 < \omega < \pi \end{cases}$$

$$h_{HT}(n) = \begin{cases} 0 & \text{for n even} \\ \dfrac{2}{n\pi} & \text{for n odd} \end{cases}$$

Ideal Differentiator

$$H_{DIF}(e^{j\omega}) = j\omega, \; 0 \le (\omega) \le \pi$$

$$h_{DIF}(n) \begin{cases} 0 & n = 0 \\ \dfrac{(-1)^n}{n} & n > 0 \end{cases}$$

Solved Examples

2. The frequency response of an FIR filter is given by the following expression.

$$H\left(e^{j\omega}\right) = e^{-j3\omega}\left[2 + 1.8\cos 3\omega + 1.2\cos 2\omega + 0.5\cos\omega\right]$$

Determine the impulse response h(n) of the filter in the form of a sequence.

Solution:

We know that for M = odd

$$H(e^{j\omega}) = e^{-j\omega\left(\frac{M-1}{2}\right)}\sum_{n=0}^{\frac{M-3}{2}} 2h(n)\cdot\cos\omega\left(\frac{M-1}{2} - n\right) + h\left(\frac{M-1}{2}\right) \quad \ldots (i)$$

Given equation is:

$$H\left(e^{j\omega}\right) = e^{-j3\omega}\left[2 + 1.8\cos 3\omega + 1.2\cos 2\omega + 0.5\cos\omega\right] \quad \ldots (ii)$$

Comparing the phase factors of (i) & (ii)

$$e^{-j\omega\left(\frac{M-1}{2}\right)} = e^{-j3\omega}$$

$$\frac{M-1}{2} = 3$$

$$M = 7$$

$$\therefore\; H(e^{j\omega}) = e^{(-j3\omega)}\left[\sum_{n=0}^{2} 2h(n)\cdot\cos\omega(3-n) + h(3)\right]$$

$$= e^{-j3\omega}\left[2h(0)\cos 3\omega + 2h(1)\cdot\cos 2\omega + 2h(2)\cdot\cos\omega + h(3)\right] \quad \ldots (iii)$$

Comparing (iii) with (ii)

$h(3) = 2$	$2h(1) = 1.2$, $h(3) = 2$
$2h(0) = 1.8$	$h(1) = 0.6$
$h(0) = 0.9$	$2h(2) = 0.5$
	$h(2) = 0.25$

$h(n) = h(m-1-n)$	$\therefore\; h(n) = \{0.9, 0.6, 0.25, 2,$
	$\uparrow$
$h(0) = h(6)$	$0.25, 0.6, 0.9\}$
$h(1) = h(5)$	
$h(2) = h(4)$	

3. Show that if z_1 is the zero of the linear phase FIR filter, then $\dfrac{1}{z_1}$ is also a zero of filter.

Solution:

We know that

$$H\left(e^{j\omega}\right) = \sum_{n=0}^{M-1} h(n)e^{-j\omega n}$$

or, $$H(z) = \sum_{n=0}^{M-1} h(z)\cdot z^n$$

For, M = odd

$$H\left(e^{j\omega}\right) = e^{-j\omega\left(\frac{M-1}{2}\right)}\left[\sum_{n=0}^{\frac{M-3}{2}} h(n)\cdot\left[e^{j\omega\left(\frac{M-1}{2}-n\right)} + e^{-j\omega\left(\frac{M-1}{2}-n\right)}\right] + h\left(\frac{M-1}{2}\right)\right]$$

or, $$H(z) = z^{-\left(\frac{M-1}{2}\right)}\left[\sum_{n=0}^{\frac{M-3}{2}} h(n)\cdot\left[z^{\frac{M-1-2n}{2}} + z^{-\frac{M-1-2n}{2}}\right] + h\left(\frac{M-1}{2}\right)\right]$$

Similarly, for M = Even

$$H\left(e^{j\omega}\right) = e^{-j\omega\left(\frac{M-1}{2}\right)}\left[\sum_{n=0}^{\frac{M-1}{2}-1} h(n)\cdot\left[e^{+j\omega\left(\frac{M-1}{2}\right)} + e^{-j\omega\left(\frac{M-1}{2}-n\right)}\right]\right]$$

$$H(z) = z^{-\frac{M-1}{2}}\left[\sum_{n=0}^{\frac{M-1}{2}} h(n)\cdot\left[z^{\frac{M-1-2n}{2}} + z^{-\left(\frac{M-1-2n}{2}\right)}\right]\right]$$

Replace 'z' by z^{-1}

$$H(z^{-1}) = z^{\frac{M-1}{2}}\left[\sum_{n=0}^{\frac{M}{2}-1}\left[z^{-\left(\frac{M-1-2n}{2}\right)} + z^{\left(\frac{M-1-2n}{2}\right)}\right]\right]$$

Multiply by $z^{-(M-1)}$

$$z^{-(M-1)}\cdot H(z^{-1}) = z^{-\left(\frac{M-1}{2}\right)}\left[\sum_{n=0}^{\frac{M}{2}-n} z^{+\frac{M-1-2n}{2}} + z^{-\frac{M-1-2n}{2}}\right]$$

Hence, thus $\boxed{H(z^{-1})\cdot z^{-(M-1)} = H(z)}$

Hence, $H(z^{-1})$ & H(z) are identical & roots of H(z) occurs onto reciprocal.

If z_1 is the zero of H(z), then $\frac{1}{z_1}$ is the zero of $H(z^{-1})$

Design of Linear Phase FIR Filter Using Window Technique

We know that desired frequency response of any digital filter is given by:

$$H_d(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h_d(n)\cdot e^{-j\omega n}$$

The coefficient $h_d(n)$ as obtained by:

$$h_d(n) = \frac{1}{2\pi}\int_{2\pi}^{\pi} H_d(e^{j\omega})\cdot e^{j\omega n}\cdot d\omega$$

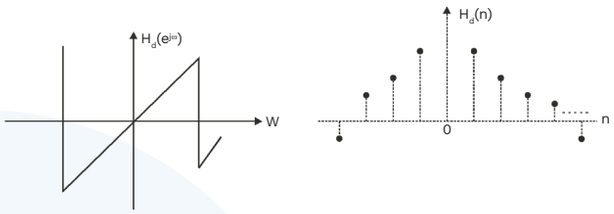


Fig. 5.13

As the impulse response $h_d(n)$ contains infinite numbers of samples, the filter is non-causal and unrealisable. To obtain the FIR filter of length M, the infinite duration impulse response just be converted to finite duration impulse response. To obtain this window function of duration 'M', i.e., $\frac{-M}{2}$ to $\frac{M}{2}$ is multiplied with $h_d(n)$, to

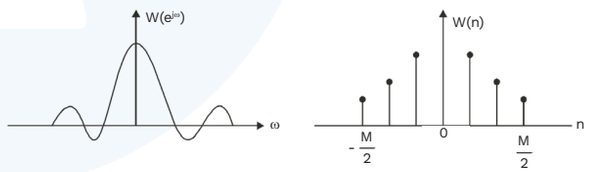


Fig. 5.14

$$\hat{h}(n) = h_d(n)\ .\ w(n)$$

$$H(e^{j\omega}) = H_d\left(e^{j\omega}\right) * W\left(e^{j\omega}\right)$$

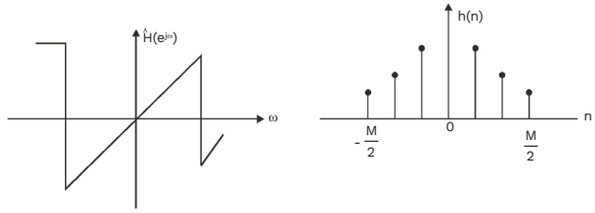


Fig. 5.15

- The causal FIR sequence is obtained by shifting h(n) right by M/2.

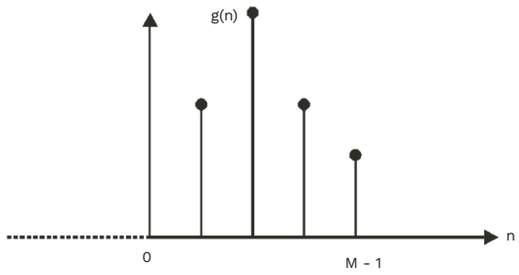


Fig. 5.16

$g(n) = h(n - M/2)$

Rectangular Windows

$$W_R(n) = 1, \; -\frac{(M-1)}{2} \le n \le \frac{M-1}{2}$$

$= 0$, elsewhere

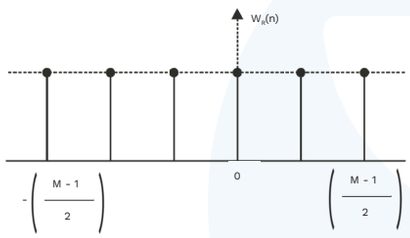


Fig. 5.17

$$W_R(e^{j\omega}) = \sum_{n=-\frac{(M-1)}{2}}^{\frac{M-1}{2}} W_R(n) e^{-j\omega n}$$

Let $n = m - \left(\frac{M-1}{2}\right)$

or, $m = n + \left(\frac{M-1}{2}\right)$

When, $n = \left(\frac{M-1}{2}\right), m = 0$

$n = \frac{M-1}{2}, m = (M-1)$

$$W_R\left(e^{j\omega}\right) = \sum_{m=0}^{M-1} e^{-j\omega\left[m-\left(\frac{M-1}{2}\right)\right]}$$

$$W_R\left(e^{j\omega}\right) = \sum_{m=0}^{M-1} e^{-j\omega m} \cdot e^{j\omega\left(\frac{M-1}{2}\right)}$$

$$W_R\left(e^{j\omega}\right) = e^{j\omega\left(\frac{M-1}{2}\right)} \sum_{m=0}^{M-1} e^{-j\omega m}$$

$$= e^{j\omega\left(\frac{M-1}{2}\right)} \left[1 + e^{-j\omega} + \ldots e^{-j\omega(M-1)}\right]$$

$$= e^{j\omega\left(\frac{M-1}{2}\right)} \left[\frac{1 - e^{-j\omega(M-1+1)}}{1 - e^{-j\omega}}\right]$$

$$= e^{j\omega\left(\frac{M-1}{2}\right)} \left[\frac{1 - e^{-j\omega M}}{1 - e^{-j\omega}}\right]$$

$$= \frac{e^{j\omega\frac{M}{2}}}{e^{j\omega/2}} \cdot \left[\frac{1 - e^{-j\omega M}}{1 - e^{-j\omega}}\right]$$

$$W_R(e^{j\omega}) = \frac{e^{j\omega\frac{M}{2}} - e^{-j\omega\frac{M}{2}}}{e^{j\omega/2} - e^{-j\omega/2}} = \frac{\sin\left(\frac{\omega M}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}$$

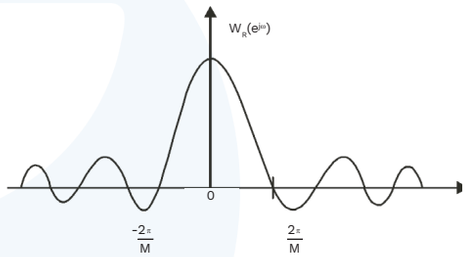


Fig. 5.18

Approximate transmition width of main lobe.

$$= \frac{4\pi}{M}$$

Magnitude response $\left|W_R\left(e^{j\omega}\right)\right| = \left|\frac{\sin\left(\frac{\omega M}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}\right|$

Phase response $\phi(\omega) = 0$

Causal Rectangular Window

$w_R(n) = 1, \; 0 \le n \le M - 1$

$$W_R\left(e^{j\omega}\right) = \sum_{n=0}^{M-1} 1 \cdot e^{-j\omega n}$$

$$\boxed{W_R\left(e^{j\omega}\right) = e^{-j\omega\left(\frac{M-1}{2}\right)} \cdot \frac{\sin\left(\frac{\omega M}{2}\right)}{\sin(\omega/2)}}$$

Magnitude response

$$\left|W_R\left(e^{j\omega}\right)\right| = \frac{\left|\sin\frac{\omega M}{2}\right|}{\left|\sin(\omega/2)\right|}, \pi \le w \le \pi$$

Phase response,

$$\phi(\omega) = -\omega\left(\frac{M-1}{2}\right); \text{ when } \sin\left(\frac{\omega M}{2}\right) \ge 0$$

$$= -\omega\left(\frac{M-1}{2}\right) + \pi \text{ ; when } \sin\left(\frac{\omega M}{2}\right) < 0$$

Solved Examples

4. A low pass filter is to be designed with the following desired frequency response.

$$H_d\left(e^{j\omega}\right) = e^{-j2\omega}, -\frac{\pi}{4} \le \omega \le \frac{\pi}{4}$$

$$= 0, \frac{\pi}{4} \le |\omega| \le \pi,$$

Determine the files coefficients $h_d(n)$, if the window function is:

$$w(n) = \begin{cases} 1 & 0 \le n \le 4 \\ 0 & \text{otherwise} \end{cases}$$

Also, determine the frequency response $H\left(e^{j\omega}\right)$ of the designed files.

Solution:
Given,

$$H_d\left(e^{j\omega}\right) = e^{-j2\omega}, -\frac{\pi}{4} \le \omega \le \frac{\pi}{4}$$

$$= 0, \frac{\pi}{4} < |\omega| \le \pi$$

We know that:

$$h_d(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d\left(e^{j\omega}\right)e^{j\omega n}\cdot d\omega$$

$$= \frac{1}{2\pi}\int_{-\pi/4}^{\pi/4} e^{-j2\omega}\cdot e^{j\omega n}\cdot d\omega$$

$$= \frac{1}{2\pi}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} e^{j\omega(n-2)}\cdot d\omega$$

$$= \frac{1}{\pi(n-2)}\cdot\left[\frac{e^{j(n-2)\cdot\pi/4} - e^{-j(n-2)\pi/4}}{2j}\right]$$

$$= \frac{1}{\pi(n-2)}\cdot\sin\frac{\pi}{4}(n-2), n \ne 2$$

$$h_d(0) = \frac{1}{2\pi} = h_d(4), h_d(1) = \frac{1}{\sqrt{2}\pi} = h_d(3)$$

$$h_d(2) = \frac{1}{4}$$

The frequency response H(ejᵠ) is given by:

$$H\left(e^{j\omega}\right) = \sum_{n=0}^{4} h(n)e^{-j\omega n}$$

$$= h(0) + h(1)e^{-j\omega} + h(2)e^{-j2\omega} + h(3)e^{-j3\omega} + h(4)e^{-j4\omega}$$

$$= h(2)e^{-j2\omega} + h(1)\left[e^{-j\omega} + e^{-j3\omega}\right] + h(0)\left[1 + e^{-j4\omega}\right]$$

$$= e^{-j2\omega}\left\{\left[h(2) + h(0)\cdot\left[e^{j2\omega} + e^{-j2\omega}\right] + h(1)\left[e^{j\omega} + e^{-j\omega}\right]\right]\right\}$$

$$= e^{-j2\omega}\left[\frac{1}{4} + \frac{1}{2\pi}\left[e^{j2\omega} + e^{-j2\omega}\right] + \frac{1}{\sqrt{2}\pi}\left[e^{j\omega} + e^{=j\omega}\right]\right]$$

$$= e^{-j2\omega}\left\{\frac{1}{4} + \frac{1}{\pi}\cos 2\omega + \frac{\sqrt{2}}{\pi}\cos\omega\right\}$$

Hamming Window Function

The causal Hamming window function is given by:

$$w_H(n) = 0.54 - 0.46\cos\frac{2\pi n}{M-1}; 0 \le n \le M-1$$

$= 0$, elsewhere

- Non-causal Hamming window is given by:

$$w_H(n) = 0.54 + 0.46\cos\frac{2\pi m}{M-1}, \text{ for } |n| \le \frac{M-1}{2}$$

$= 0$, elsewhere

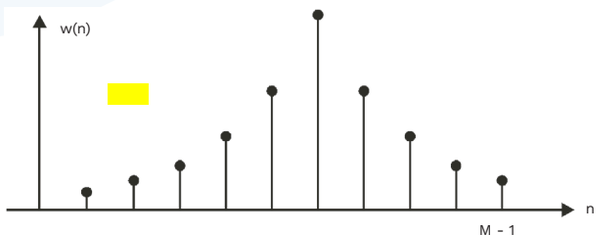


Fig. 5.19

Hanning Window Function

A causal Hanning window is given by:

$$w_{Hann}(n) = 0.5 - 0.5\cos\frac{2\pi n}{M-1}, 0 \le n \le M-1$$

$= 0$, elsewhere

- A non-causal Hanning window is given by:

$$w_{Hann}(n) = 0.5 + 0.5\cos\pi\frac{2\pi n}{M-1}, |n| \le \frac{M-1}{2}$$

$$= 0, \quad \text{elsewhere}$$

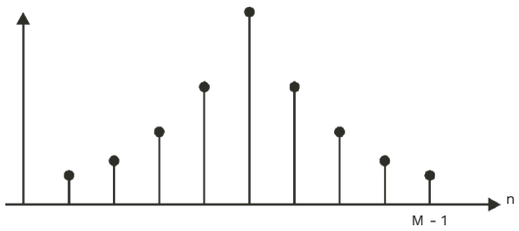


Fig. 5.20

Blackman Window

A causal Blackman window is given by:

$$w_B(n) = 0.42 - 0.5\cos\frac{2\pi n}{M-1} + 0.08\cos\frac{4\pi n}{M-1},$$

$$0 \le n \le M-1$$

$$= 0, \quad \text{elsewhere}$$

A non-causal Blackman window is given by:

$$w_B(n) = 0.42 + 0.5\cos\frac{2\pi n}{M-1} + 0.08\cos\frac{4\pi n}{M-1},$$

$$\text{for } |n| \le \frac{M-1}{2}$$

$$= 0, \quad \text{elsewhere}$$

Bartlett Window

A causal Bartlett window is given by:

$$w_{Bart}(n) = \begin{cases} 1 - \dfrac{2\left|n - \dfrac{(M-1)}{2}\right|}{M-1}, & 0 \le n \le M-1 \\ 0, & \text{otherwise} \end{cases}$$

- A non-causal Bartlett window is given by:

$$w_{Bart}(n) = \begin{cases} 1 - \dfrac{2|n|}{M-1}, & \text{for } |n| \le \dfrac{(M-1)}{2} \\ 0, & \text{else where} \end{cases}$$

Solved Examples

5. The desired frequency response of a low pass filter is given by:

$$H_d(e^{j\omega}) = e^{-j3\omega}, -\frac{3\pi}{4} \le \omega \le \frac{3\pi}{4}$$

$$= 0, \quad \frac{3\pi}{4} < |\omega| \le \pi$$

Determine $H(ej^{\omega})$ for m = 7, using Hamming window.

Solution:

The filter coefficient is given by:

$$h_d(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(e^{j\omega})e^{j\omega n}\cdot d\omega$$

$$= \frac{1}{2\pi}\int_{-\frac{3\pi}{4}}^{\frac{3\pi}{4}} e^{-j3\omega}\cdot e^{j\omega n}\cdot d\omega$$

$$h_d(n) = \frac{\sin 3\pi(n-3)/4}{\pi(n-3)}, n \ne 3$$

$h_d(0) = 0.0750$ $\quad$ $h_d(4) = 0.2251$

$h_d(1) = -0.1592$ $\quad$ $h_d(5) = -0.1592$

$h_d(2) = 0.2251$ $\quad$ $h_d(6) = 0.0750$

$h_d(3) = 0.75$

Hamming window function is given by:

$$w(n) = 0.54 - 0.46\cos\frac{2\pi n}{M-1}; 0 \le n \le M-1$$

$$= 0, \text{ elsewhere}$$

$w(0) = 0.08$ $\quad$ $w(3) = 1$

$w(1) = 0.31$ $\quad$ $w(4) = 0.77$

$w(2) = 0.77$ $\quad$ $w(5) = 0.31$

$h(n) = h_d(n)\, w(n)$ $\quad$ $w(6) = 0.08$

$h(0) = 0.006$ $\quad$ $h(2) = 0.1733$

$h(4) = 0.1733$

$\because$ $h(1) = -0.0494$ $\quad$ $h(3) = 0.75$

$h(5) = -0.0494$ $\quad$ $h(6) = 0.006$

6. Design the symmetry FIR low pass filter for which desired frequency response is:

$$H_d(\omega) = e^{-j\omega\tau}, |\omega| \le \omega_c$$

$$= 0, \text{ elsewhere}$$

The length of the filter should be 7 and $w_c = 1$ rad/s using Hanning window function.

Solution:

Given,

$$H_d(\omega) = e^{-j\omega\tau}, |\omega| \le \omega_c$$

$$H_d(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)\cdot e^{j\omega n}\cdot d\omega$$

$$h_d(n) = \frac{1}{2\pi}\int_{-1}^{1} e^{-j\omega\tau}\cdot e^{j\omega n}.d\omega$$

or, $h_d(n) = \frac{1}{2\pi}\int_{-1}^{1} e^{j\omega(n-\tau)} \cdot d\omega$

$$= \frac{1}{2\pi} \cdot \left[\frac{e^{j\omega(n-\tau)}}{j(n-\tau)}\right]_{-1}^{1}$$

$$= \frac{\sin(n-\tau)}{\pi(n-\tau)} \text{ for } n \neq \tau$$

for, $n = \tau$

$$h_d(n) = \frac{1}{\pi}$$

$$h_d(n) = \begin{cases} \frac{1}{\pi}\frac{\sin(n-\tau)}{(n-\tau)} = \frac{1}{\pi}\frac{\sin\left\{n-\left(\frac{M-1}{2}\right)\right\}}{n-\left(\frac{M-1}{2}\right)} \\ \frac{1}{\pi} \qquad\qquad n = \tau \end{cases}$$

$\because$ FIR filter is symmetry,

$\therefore$ $h_d(n) = h_d(M-1-n)$

$$h_d(0) = h_d(6) = \frac{\sin(-3)}{-3\pi} = 0.01497$$

$$h_d(1) = h_d(5) = \frac{\sin(-2)}{-2\pi} = 0.14472$$

$$h_d(2) = h_d(4) = \frac{\sin(-1)}{\pi} = 0.26785$$

and $h_d(3) = \frac{1}{\pi} = 0.31838$

We know that Hanning window function,

$$w(n) = \frac{1}{2}\left[1-\cos\frac{2\pi n}{M-1}\right]$$

$$w(0) = \frac{1}{2}[1-\cos 0] = 0$$

$$w(1) = \frac{1}{2}\left[1-\cos\left[\frac{2\pi}{6}\right]\right] = \frac{1}{4}$$

$$w(2) = \frac{1}{2}\left[1-\cos\left[\left(\frac{4\pi}{6}\right)\right]\right] = \frac{3}{4}$$

$$w(3) = \frac{1}{2}\left[1-\cos\left[\left(\frac{6\pi}{6}\right)\right]\right] = 1$$

$$w(4) = \frac{1}{2}\left[1-\cos\left[\left(\frac{8\pi}{6}\right)\right]\right] = \frac{3}{4}$$

$$w(5) = \frac{1}{2}\left[1-\cos\left[\left(\frac{10\pi}{6}\right)\right]\right] = \frac{1}{4}$$

$$w(6) = \frac{1}{2}\left[1-\cos\left[\left(\frac{12\pi}{6}\right)\right]\right] = 0$$

The desired frequency response is:

$h(n) = h_d(n) \cdot w(n)$; for $0 \le n \le 6$

$\therefore$ $h(0) = h_d(0) \cdot w(0) = 0 \quad = h(6)$

$h(1) = h_d(1) \cdot w(1) = 0.03618 = h(5)$

$h(2) = h_d(2)\, w(2) = 0.20089 = h(6)$

$h(3) = h_d(3)\, w(3) = 0.31831$

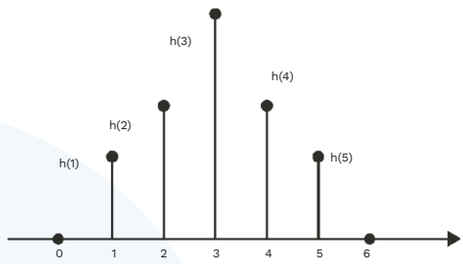


Fig. 5.21

7. An FIR linear phase filter has the following impulse response.

$h_d(n) = 1, \qquad 0 \le n \le 4$

$= 0, \qquad$ elsewhere

Use Bartlett's window to compute the impulse response of the filter. Find its magnitude & phase response as a function of frequency.

Solution:

Given, $h_d(n) = 1, \qquad 0 \le n \le 4$

Bartlett's window is given by:

$$w_{Bart} = 1 - \frac{2\left|n-\frac{M-1}{2}\right|}{M-1} ; 0 \le n \le M-1$$

$$= 0 , \qquad \text{otherwise}$$

$$w_{Bart}(n) = \frac{1-2\left|n-\frac{4}{2}\right|}{5} = \frac{1-2|n-2|}{5}$$

$\because$ $h(n) = h_d(n) \cdot w_{Bart}(n)$

$h(0) = h_d(0) \cdot w_{Bart}(0) = 1$

$$= 1 \times \left[1-\frac{2|0-2|}{4}\right] = 0$$

$$h(1) = 1\times\left[1-\frac{2|1-2|}{4}\right] = 1-\frac{1}{2} = 0.5$$

$$h(2) = 1\times\left[1-\frac{2|2-2|}{4}\right] = 1$$

As FIR filter has a linear phase,

$h(0) = h(4) = 0, \quad h(1) = h(3) = 0.5, \quad h(2) = 1$

$$H\left(e^{j\omega}\right) = \sum_{n=0}^{4} h(n)\cdot e^{-j\omega n}$$

$$= h(0) + h(1)e^{-j\omega} + h(2)\,e^{-j2\omega} + h(3)\cdot e^{-j3\omega} + h(4)e^{-j4\omega}$$

$$= h(2)e^{-j2\omega} + h(0)\left[1+e^{-j4\omega}\right] + h(1)\left[e^{-j\omega}+e^{-j3\omega}\right]$$

$$= e^{-j2\omega}\left[h(2) + h(0)\left[e^{j2\omega}+e^{-j2\omega}\right] + h(1)\left[e^{j\omega}+e^{-j\omega}\right]\right]$$

$$= e^{-j2\omega}\left[h(2) + 2h(0)\cdot\cos 2\omega + 2h(1)\cdot\cos\omega\right]$$

$$H\left(e^{j\omega}\right) = e^{-j2\omega}\left[1+\cos\omega\right]$$

Magnitude Response

$$\left|H\left(e^{j\omega}\right)\right| = \left|1+\cos\omega\right|$$

ω	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
$\left\|H(e^{j\omega})\right\|$	2	1.866	1.707	1.5	1

$2\pi/3$	$3\pi/4$	$5\pi/6$	π
0.5	0.293	0.134	0

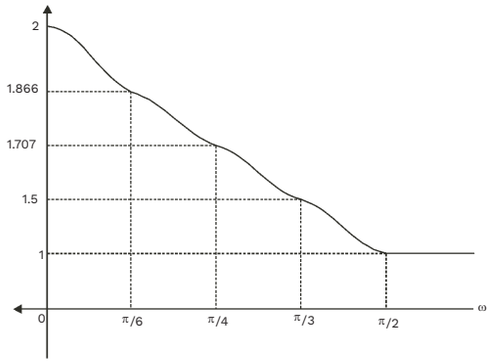


Fig. 5.22

Phase Response

ω	$-\pi$	$-\pi/2$	0	$\pi/2$	π
$\phi(\omega) = -2\omega$	2π	π	0	$-\pi$	-2π

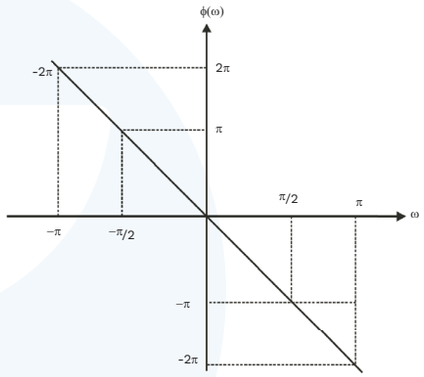


Fig. 5.23

Important Frequency Domain Characteristics of Some Window Functions

Type of window	Approximate transition width of main lobe	Peak Amplitude of side loab(dB)	Minimum stop Band Attentation (dB)	Transition Band width $\Delta\omega$
Rectangular	$\frac{4\pi}{M}$	–13	–21	$\frac{0.92\pi}{M}$
Bartlett	$\frac{8\pi}{M}$	–25	–25	$\frac{3.10\,\pi}{M}$
Hanning	$\frac{8\pi}{M}$	–31	–44	$\frac{3.11\pi}{M}$
Hamming	$\frac{8\pi}{M}$	–41	–53	$\frac{3.32\pi}{M}$
Blackman	$\frac{12\pi}{M}$	–57	–74	$\frac{5.56\pi}{M}$

Table 5.2

Kaiser Window

The causal Kaiser window, which can be used for both odd and even values of M, given by:

$$w_k(n) = \begin{cases} \dfrac{I_0\left[\beta\sqrt{1-\left(1-\dfrac{2n}{M-1}\right)^2}\right]}{I_0(\beta)} & 0 \le n \le M-1 \\ 0 & \text{otherwise} \end{cases}$$

Variable parameter β controls the peak side lobe level.

The non-causal Kaiser window, which is used for odd values of M is given as:

$$w_k(n) = \begin{cases} \dfrac{I_0\left[\beta\sqrt{1-\left(\dfrac{2n}{M-1}\right)^2}\right]}{I_0(\beta)}, & -\dfrac{M-1}{2} \le n \le \dfrac{M-1}{2} \\ 0 & \text{otherwise} \end{cases}$$

$I_0(\beta) \rightarrow$ Zero order magnified Bessed function.

Design of IIR Digital Filters from Analog Filters

The traditional approach to the design IIR digital filters involves the transformation of an analog filter into a digital filter meeting prescribed specifications.

1. The art of analog filter design is highly advanced. So, useful result can be achieved. It is advantageous to utilise the design procedure already developed for analog filters.
2. Many useful analog design methods have relatively simple closed from design formula therefore, digital filter design methods based on such analog design formula are rather simple to implement.
3. In many applications it is interest to use a digital filter to simulate the performance of an analog linear time invariant filter.

An analog filter can be described by its system function

$$H_a(s) = \frac{B(s)}{A(s)} = \frac{\sum_{k=0}^{M}\beta_k s^k}{\sum_{k=0}^{N}\alpha_k s^k} \quad \ldots (i)$$

where $\{\alpha_k\}$ and $\{\beta_k\}$ are the filter coefficients.

$$H_a(s) = \int_{-\infty}^{\infty} h(t)\, e^{-st} dt \quad \ldots (ii)$$

Alternatively the analog filter having the rational system function H(s), given in equation (i) can be described by the linear constant coefficient differential equation.

$$\sum_{k=0}^{N}\alpha_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M}\beta_k \frac{d^k x(t)}{dt^k}$$

The LTI system with system function H(s) is stable if all its poles lie in the left half of the s-plane.

Consequently, if the conversion technique is to be effective, it should possess the following desirable properties.

1. The $j\Omega$ axis in the s-plane should map into the unit circle in the z-plane.
2. The Left Half Plane (LHP) of the s-plane should map into inside of the unit circle in the z-plane. Thus, a stable analog filter will be converted to a stable digital filter.

IIR Filter Design by Approximation of Derivative

$$\left.\frac{dy(t)}{dt}\right|_{t=nT} = \frac{y(nT) - y(nT-T)}{T}$$

$$= \frac{y(n) - y(n-1)}{T}$$

When T represents the sampling interval,

$y(n) = y(nT)$

The analog differentiator with output $\frac{dy(t)}{dt}$ has the system function H(s) = s, while the digital system that produces the output [y(n) – y(n – 1)]/T has the system function $H(z) = \frac{1-z^{-1}}{T}$.

$$\boxed{s = \frac{1-z^{-1}}{T}}$$

$$\left.\frac{d^2y(t)}{dt^2}\right|_{t=nt} = \frac{d}{dt}\left[\frac{y(nT) - y(nT-T)}{T}\right]_{t=nT}$$

$$= \frac{[y(nT) - y(nT-T)]/T - [y(nT-T) - y(nT-2T)]/T}{T}$$

$$= \frac{y(n) - 2y(n-1) + y(n-2)}{T^2}$$

$$s^2 = \left(\frac{1-z^{-1}}{T}\right)^2$$

System Function for Digital IIR Filter is

$$H(z) = H_a(s)\Big|_{s=\frac{1-z^{-1}}{T}}$$

where $H_a(s)$ is system function of analog filter.

$$s = \frac{1-z^{-1}}{T}$$

$$z = \frac{1}{1-sT}$$

$$z = \frac{1}{1-j\Omega T}$$

$$z = \frac{1}{1+\Omega^2T^2} + j\frac{\Omega T}{1+\Omega^2T^2}$$

Ω varies from $-\infty$ to ∞, the corresponding locus of point in the z-plane is a circle of radius 1/2 and with the centre at $z = \frac{1}{2}$.

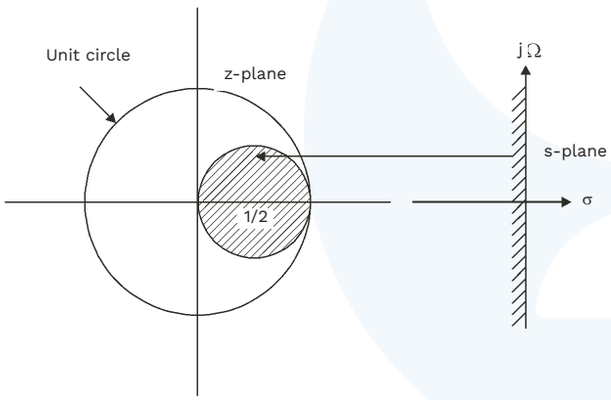


Fig. 5.24

Mapping from S-plane to Z-plane

1. Points in the Left Hand Plane (LHP) of s-plane are mapped into corresponding point inside this circle in the z-plane.
2. Points in the Right Hand Plane (RHP) of the s-plane are mapped into points outside this circle.
3. This $j\Omega$ axis maps on to the perimeter of the circle of radius 1/2 in z-plane.

$$s = \frac{1-z^{-1}}{T}$$

$$z = \frac{1}{1-sT}$$

$$= \frac{0.5 - 0.5sT + 0.5 + 0.5sT}{1-sT}$$

$$= \frac{0.5(1-sT) + 0.5(1+sT)}{1-sT}$$

$$= 0.5 + \frac{0.5(1+sT)}{(1-sT)}$$

$$z - 0.5 = 0.5\frac{(1+sT)}{(1-sT)} = 0.5\frac{[1+(\sigma+j\Omega)T)]}{[1-(\sigma+j\Omega)T]}$$

To find where $j\Omega$ axis maps, $\sigma = 0$

$$z - 0.5 = 0.5\frac{(1+j\Omega T)}{(1-j\Omega T)}$$

$$\boxed{|z-0.5| = 0.5}$$

$j\Omega$ axis is mapped into a circle of radius 0.5 and centred at z = 0.5

Consequently, this mapping has the desirable property that a stable analog filter is transformed into a stable digital filter.

Solved Examples

8. Convert the analog band pass filter with system function.

$$H_a(s) = \frac{1}{(s+0.1)^2+9}$$

into a digital IIR filter by use of the backward difference for the derivative.

Solution:

Put,

$$s = \frac{1-z^{-1}}{T}$$

$$H(z) = \frac{1}{\left(\frac{1-z^{-1}}{T} + 0.1\right)^2 + 9}$$

$$H(z) = \frac{T^2\left|(1+0.2T) + 9.01\,T^2\right.}{1 - \frac{2(1+01)}{1+0.2T+9.01T^2}z^{-1} + \frac{1}{1+0.2T+9.01T^2}z^{-2}}$$

IIR Filter Design by Impulse Invariance

An IIR filter having a unit sample response h(n) that is the sampled version of the impulse response of the analog filter, i.e.,

$h(n) \equiv h(nT), \qquad n = 0, 1, 2, \text{----------}$

Where, T is sampling interval.

Expressed in the context of sampling the impulse response of an analog filter with frequency response $H_a(f)$, the digital filter with unit sample response $h(n) \equiv h_a(nT)$ has frequency response.

$$H(f) = F_s \sum_{k=-\infty}^{\infty} H_a\left[(f-k)F_s\right]$$

$$\text{or } H(\omega) = F_s \sum_{k=-\infty}^{\infty} H_a\left[(\omega - 2\pi k)F_s\right]$$

$$\boxed{H(\Omega T) = \frac{1}{T}\sum_{k=-\infty}^{\infty} H_a\left(\Omega - \frac{2\pi k}{T}\right)}$$

Note: Impulse Invariance method is inappropriate for designing high pass filter due to the spectrum aliasing that results from the sampling process.

$$H(z)\big|_{z=e^{sT}} = \frac{1}{T}\sum_{k=-\infty}^{\infty} H_a\left(s - j\frac{2\pi k}{T}\right)$$

$$H(z) = \sum_{n=0}^{\infty} h(n)z^{-n}$$

$$H(z)\big|_{z=e^{sT}} = \sum_{n=0}^{\infty} h(n)e^{-sTn}$$

$$z = e^{sT}$$

$$re^{j\omega} = e^{(\sigma + j\Omega)T}$$

$$re^{j\omega} = e^{\sigma T}e^{j\Omega T}$$

$$r = e^{\sigma T}$$

$$\omega = \Omega T$$

$\sigma < 0 \quad \Rightarrow \quad 0 < r < 1$

$\sigma > 0 \quad \Rightarrow \quad r > 1$

$\sigma = 0 \quad \Rightarrow \quad r = 1$

$H_a(s)$ (Analog filter) $\rightarrow$ $H(z)$

$\downarrow$ $\uparrow z$

$h_a(t) \rightarrow h_a(nT) = h(n)$

h(n) is invariant,

$$\frac{1}{s - p_k} \rightarrow \frac{1}{s - e^{p_k T}z^{-1}}$$

Therefore, LHP in the s-plane is mapped inside the unit circle in the z-plane, and RHP in the s-plane is mapped outside the unit circle in the z-plane.

$j\Omega$ axis is mapped into the unit circle in z-plane.

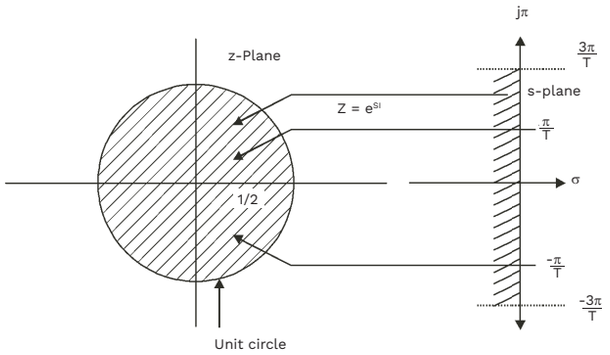


Fig. 5.25

$$H_a(s) = \sum_{k=1}^{N} \frac{c_k}{s - p_k}$$

$\{p_k\} \Rightarrow$ Poles of analog filter

$\{c_k\} \Rightarrow$ Coefficients in partial fraction expansion

$$h_a(t) = \sum_{k=1}^{N} c_k e^{p_k t} \; ; t \geq 0$$

If we sample periodically at t = nT, we have:

$$h(n) = h_a(nT)$$

$$= \sum_{k=1}^{N} c_k e^{p_k Tn}$$

$$H(z) = \sum_{n=0}^{\infty} h(n)z^{-n}$$

$$= \sum_{n=0}^{\infty}\left(\sum_{k=1}^{N} c_k e^{p_k Tn}\right)z^{-n}$$

$$= \sum_{k=1}^{N} c_k \sum_{n=0}^{\infty}\left(e^{p_k T}z^{-1}\right)^n$$

$$\boxed{H(z) = \sum_{k=1}^{N} \frac{c_k}{1 - e^{p_k T}z^{-1}}}$$

Poles of digital filter, $z_k = e^{p_k T}$, k = 1, 2, ... N

Solved Examples

9. Convert the analog filter with system function

$$H_a(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

into a digital IIR filter by means of the Impulse Invariance method.

Solution:

Poles, $p_k = -0.1 \pm j3$

Zero, $s = -0.1$

Pole zero location for analog filter:

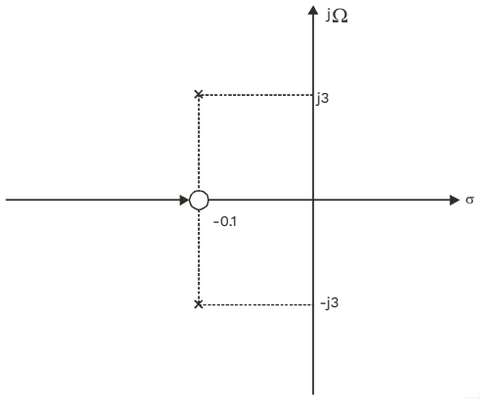


Fig. 5.26

$$H(s) = \frac{1/2}{s+0.1-j3} + \frac{1/2}{s+0.1+j3}$$

$$H(z) = \frac{1/2}{1-e^{-0.1T}e^{j3T}z^{-1}} + \frac{1/2}{1-e^{-0.1T}e^{-j3T}z^{-1}}$$

$$\boxed{H(z) = \frac{1-(e^{-0.1T}\cos 3T)z^{-1}}{1-(2e^{-0.1T}\cos 3T)z^{-1}+e^{-0.2T}z^{-1}}}$$

10. Transform the analog filter transfer function

$$H_a(s) = \frac{4s+7}{s^2+5s+4}$$

into a digital filter H(z) using the impulse invariance method at $F_s = 2Hz$.

Solution:

$$H_a(s) = \frac{4s+7}{s^2+5s+4} = \frac{3}{(s+4)} + \frac{1}{(s+3)}$$

$$\frac{1}{s+p_k} \longleftrightarrow \frac{1}{1-e^{-p_kT}z^{-1}}$$

$$T = \frac{1}{F_s} = 0.5$$

$$\frac{1}{s+4} \longleftrightarrow \frac{1}{1-e^{-4T}z^{-1}} = \frac{1}{1-e^{-2}z^{-1}}$$

$$\frac{1}{s+1} \longleftrightarrow \frac{1}{1-e^{-T}z^{-1}} = \frac{1}{1-e^{-0.5}z^{-1}}$$

$$H(z) = \frac{3}{1-e^{-2}z^{-1}} + \frac{1}{1-e^{-0.5}z^{-1}}$$

$$\boxed{H(z) = \frac{4-1.9549z^{-1}}{1-0.7419z^{-1}+0.0821z^{-2}}}$$

11. Transform the analog filter transfer function

$$H_a(s) = \frac{1}{(s+a)^2}$$

into a digital filter H(z) using the impulse invariant method.

Solution:

$$H_a(s) = \frac{1}{(s+a)^2}$$

$$h_a(t) = te^{-at}u(t)$$

$$h_a(t)\big|_{t=nT} = h_a(n\tau) = h(n) = nTe^{-an\tau}u(nT)$$

$$h(n) = Tn(e^{-aT}z^{-1})^n u(n)$$

$$\boxed{H(z) = \frac{Te^{-aT}z^{-1}}{\left(1-e^{-aT}z^{-1}\right)^2}}$$

Advantages of Impulse Invariance Method

1. It preserves the stability of the analog filters.
2. The relationship between analog and digital frequencies are linear, i.e., $\omega = \Omega T$.
3. Since the relationship between analog and digital frequencies is linear, the specification for a digital filter can be easily transformed into those for an analog filter.

Disadvantages of Impulse Invariance Method

1. $z = e^{sT}$ does not describe one-to-one mapping between the s-plane and z-plane. Impulse invariance mapping is many to one mapping.
2. As no analog filter of finite order can be exactly band limited, some aliasing will occur.
3. This method is not suitable for designing HPF and BSF due to spectrum aliasing resulting from the sampling process.

IIR Filter Design by Bilinear Transformation

The IIR filter design using (i) approximation of derivatives methods and (ii) the impulse invariant method are appropriate for the design of low pass filter and band pass filter whose resonant frequencies are low. These techniques are not suitable for high pass or band reject filters. This limitation is overcome in the mapping technique called bilinear transformation.

Let the system function of the analog filter be:

$$H(s) = \frac{b}{s+a}$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b}{s+a}$$

$$sY(s) + aY(s) = bX(s)$$

$$\frac{dy(t)}{dt} + ay(t) = bx(t)$$

$$\int_{nT-T}^{nT} \frac{dy(t)}{dt}dt + a\int_{nT-T}^{nT} y(t)dt = b\int_{nT-T}^{nT} x(t)dt$$

Trapezoidal rule for numeric integration given by:

$$\int_{nT-T}^{nT} a(t)dt = \frac{T}{2}\left[a(nT) + a(nT-T)\right]$$

$$\int_{nT-T}^{nT} dy(t) + a\int_{nT-T}^{nT} y(t)dt \pm b\int_{nT-T}^{nT} x(t)dt$$

$$y(nT) - y(nT-T) + \frac{aT}{2}\left[y(nT) + y(nT-T)\right]$$

$$= \frac{bT}{2}\left[x(nT) + x(nT-T)\right]$$

$$y(n) \equiv y(nT)$$

$$x(n) \equiv x(nT)$$

$$y(n) - y(n-1) + \frac{aT}{2}\left[y(n) + y(n-1)\right]$$

$$= \frac{bT}{2}\left[x(n) + x(n-1)\right]$$

Taking z transform on both sides:

$$Y(z) - z^{-1}Y(z) + \frac{aT}{2}\left[Y(z) + z^{-1}Y(z)\right]$$

$$= \frac{bT}{2}\left[X(z) + z^{-1}X(z)\right]$$

$$\left(1+\frac{aT}{2}\right)Y(z) - \left(1-\frac{aT}{2}\right)z^{-1}Y(z) = \frac{bT}{2}\left(1+z^{-1}\right)X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{(bT/2)\left(1+z^{-1}\right)}{\left(1+\frac{aT}{2}\right) - \left(1-\frac{aT}{2}\right)z^{-1}}$$

$$H(z) = \frac{b}{\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) + a}$$

$$H(s) = \frac{b}{s+a}$$

Clearly, mapping from s-plane to the z-plane is:

$$\boxed{s = \frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)}$$

This is called bilinear transformation.

$$z = re^{j\omega}$$

$$s = \sigma + j\Omega$$

$$s = \frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) = \frac{2}{T}\left(\frac{z-1}{z+1}\right)$$

$$= \frac{2}{T}\left(\frac{re^{j\omega}-1}{re^{j\omega}+1}\right) = \frac{2}{T}\left[\frac{r\cos\omega + jr\sin\omega - 1}{r\cos\omega + jr\sin\omega + 1}\right]$$

$$\left[\frac{r\cos\omega + 1 - jr\sin\omega}{r\cos\omega + 1 - jr\sin\omega}\right]$$

$$= \frac{2}{T}\left(\frac{r^2-1}{1+r^2+2r\cos\omega} + j\frac{2r\sin\omega}{1+r^2+2r\cos\omega}\right)$$

Consequently,

$$\sigma = \frac{2}{T}\left(\frac{r^2-1}{1+r^2+2r\cos\omega}\right)$$

$$\Omega = \frac{2}{T}\left(\frac{2r\sin\omega}{1+r^2+2r\cos\omega}\right)$$

if $r < 1$, then $\sigma < 0$

if $r > 1$, then $\sigma > 0$

Consequently, LHP in s-plane maps into the inside of the unit circle in z-plane and RHP in s-plane maps into the outside of the circle.

Where, $r = 1$, and $\sigma = 0$

$$\Omega = \frac{2}{T}\frac{\sin\omega}{1+\cos\omega} = \frac{2}{T}\left[\frac{2\sin\left(\frac{\omega}{2}\right)\cos\left(\frac{\omega}{2}\right)}{2\cos^2\left(\frac{\omega}{2}\right)}\right]$$

$$\Omega = \frac{2}{T}\tan\left(\frac{\omega}{2}\right)$$

$$\boxed{\omega = 2\tan^{-1}\left(\frac{\Omega T}{2}\right)}$$

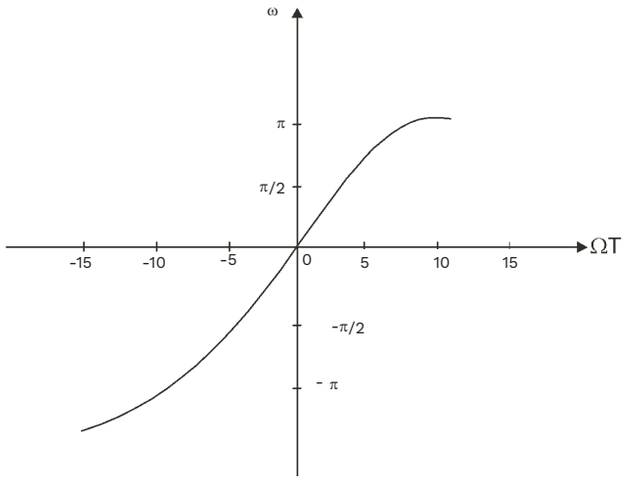


Fig. 5.27

- Entire range in Ω is mapped only once into the range $-\pi \le \omega \le \pi$. However, mapping is highly non-linear.
- We observe that frequency compression or frequency warping is due to non-linearly of the arctangent function.
- Bilinear transformation maps the point $s = \infty$ into the point $z = -1$.

$$H(s) = \frac{b}{s+a}$$

The above system function represents a low pass filter; which has a zero at $s = \infty$. This results in a digital filter that has a zero at $z = -1$.

Solved Examples

12. Convert the analog filter with system function

$$H_a(s) = \frac{s+0.1}{(s+0.1)^2+16}$$

into a digital IIR filter by means of bilinear transformation. The digital filter is to have a resonant frequency of $\omega_r = \frac{\pi}{2}$.

Solution:

$$\omega_r = \frac{\pi}{2}$$

$$T = \frac{1}{2}$$

$$s = \frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)$$

$$s = 4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)$$

Using bilinear transformation:

$$H(z) = H(s)\Big|_{S=4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)}$$

$$= \frac{4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+0.1}{\left[4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+0.1\right]^2+16}$$

$$\boxed{H(z) = \frac{0.128+0.006z^{-1}-0.122z^{-2}}{1+0.0006z^{-1}+0.975z^{-2}}}$$

Coefficient of z^{-1} is very small in denominator of H(z) and can be approximated by zero.

$$H(z) = \frac{0.128+0.006z^{-1}-0.122z^{-2}}{1+0.975z^{-2}}$$

Poles of filter, $P_{1,2} = 0.987\, e^{\pm j\frac{\pi}{2}}$

Zeros of filter, $z_{1,2} = -1, 0.95$

13. Design a single pole low pass digital filter bilinear transformation applied to the analog filter.

$$H(s) = \frac{\Omega_c}{s+\Omega_c}$$

Where, Ω_c is the 3dB band width of the analog filter.

Solution:

$$\omega_c = 0.2\pi$$

$$\Omega_c = \frac{2}{T}\tan\left(\frac{\omega_c}{2}\right)$$

$$\Omega_c = \frac{2}{T}\tan(0.1\,\pi) = \frac{0.65}{T}$$

$$s = \frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)$$

$$H(z) = \frac{\frac{0.65}{T}}{\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+\frac{0.65}{T}} = \frac{0.65(1+z^{-1})}{2-2z^{-1}+0.65+0.65\,z^{-1}}$$

$$= \frac{0.65(1+z^{-1})}{2.65-1.35z^{-1}} = \frac{0.245(1+z^{-1})}{1-0.509z^{-1}}$$

$$H(z) = \frac{0.245(1+z^{-1})}{1-0.509z^{-1}}$$

$$H(w) = \frac{0.245\left(1+e^{-j\omega}\right)}{1-0.509\,e^{-j\omega}}$$

$H(0) = 1$

$|H(0.2\,\pi)| = 0.707$

14. Using bilinear transformation to obtain H(z) if $H(s) = \frac{1}{(s+1)^2}$, and T = 0.15

Solution:

$$s = \frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) = \frac{2}{0.1}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) = 20\left(\frac{1-z^{-1}}{1+z^{-1}}\right)$$

$$H(z) = \frac{1}{\left[20\left(\frac{1-z^{-1}}{1-z^{-1}}\right)+1\right]^2}$$

$$\boxed{H(z) = \frac{0.0476(1+z^{-1})^2}{\left(1-0.9048z^{-1}\right)^2}}$$

15. Using bilinear transformation to obtain H(z) if $H(s) = \frac{2}{(s+1)(s+3)}$, and T = 0.1s

Solution:

$$s = \frac{2}{T}\left(\frac{1-z^{1}}{1+z^{-1}}\right) = \frac{2}{0.1}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)$$

$$s = 20\left(\frac{1-z^{-1}}{1+z^{-1}}\right)$$

$$H(z) = \frac{2}{\left[20\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+1\right]\left[20\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+3\right]}$$

$$\boxed{H(z) = \frac{0.0041(1+z^{-1})^2}{1-1.644\,z^{-1}+0.668\,z^{-2}}}$$

16. Convert the analog filter into a digital filter whose system function is:

$$H(s) = \frac{s+0.2}{(s+0.2)^2+9}$$

Use the impulse invariant technique. Assume t = 1s.

Solution:

The system response of the analog filter is of the standard form

$$H(s) = \frac{s+a}{(s+a)^2+b^2}$$

Where, a = 0.2 and b = 3. The system response of the digital filter can be obtained using following equation.

$$H(z) = \frac{1-e^{-aT}(\cos bT)z^{-1}}{1-2e^{-aT}(\cos bT)z^{-1}+e^{-2aT}z^{-2}}$$

$$= \frac{1-e^{-0.2T}(\cos 3T)z^{-1}}{1-2e^{-0.2T}(\cos 3T)z^{-1}+e^{-0.4T}z^{-2}}$$

T = 1 s

$$H(z) = \frac{1-(0.8187)(-0.99)z^{-1}}{1-2(0.8187)(-0.99)z^{-1}+0.6703z^{-2}}$$

$$H(z) = \frac{1+(0.8105)z^{-1}}{1+1.6210\,z^{-1}+0.6703z^{-2}}$$

17. For analog transfer function

$$H(s) = \frac{1}{(s+1)(s+2)}$$

Determine H(z) using impulse invariant technique. Assume T = 1s.

Solution:

$$H(s) = \frac{1}{(s+1)(s+2)} = \frac{1}{s+1} - \frac{1}{s+2}$$

T = 1 sec

$$H(z) = \frac{1}{1-e^{-T}z^{-1}} - \frac{1}{1-e^{-2T}z^{-1}}$$

$$= \frac{z^{-1}[e^{-T}-e^{-2T}]}{1-(e^{-T}+e^{-2T})z^{-1}+e^{-3T}z^{-2}}$$

$$H(z) = \frac{0.2326z^{-1}}{1-0.5032z^{-1}+0.0498z^{-2}}$$

Advantages of Bilinear Transformation

1. The bilinear transformation method is a stable and one-to-one mapping method. There is a one-to-one relationship between Ω and ω.
2. There is no aliasing
3. There are no limitation on the use of bilinear transformation. It is adequate for all filter types.
4. Bilinear transformation preserves the magnitude response of an analog filter with piece-wise constant magnitude.

Disadvantages of Bilinear Transformation Method

1. The non-linear relationship between the analog frequency (Ω) and digital frequency (ω) results in frequency warping.
2. A linear phase analog filter cannot be transformed to a linear phase digital filter using bilinear transformation.

Comparison between Impulse Invariant Method and Bilinear Transformation

Impulse Invariant Method	Bilinear Transformation Method
It is many-to-one mapping.	It is one-to-one mapping.
The relation between analog frequency Ω and digital frequency ω is linear, given by $\omega = \Omega\tau$.	The relation between analog frequency Ω and digital frequency ω is non-linear, given by $\omega = 2\tan^{-1}\left(\frac{\Omega T}{2}\right)$.
This method suffers from aliasing.	There is no aliasing.
It is useful for LPF and BPF, whose frequency response is band limited.	It is useful for all type of filters.
At high sampling frequencies or low sampling interval T, the magnitude response and phase response of the analog filter can be preserved.	The magnitude response of the analog filter remains unchanged due to frequency war ping the phase response of the analog filter cannot be preserved.

Table 5.3

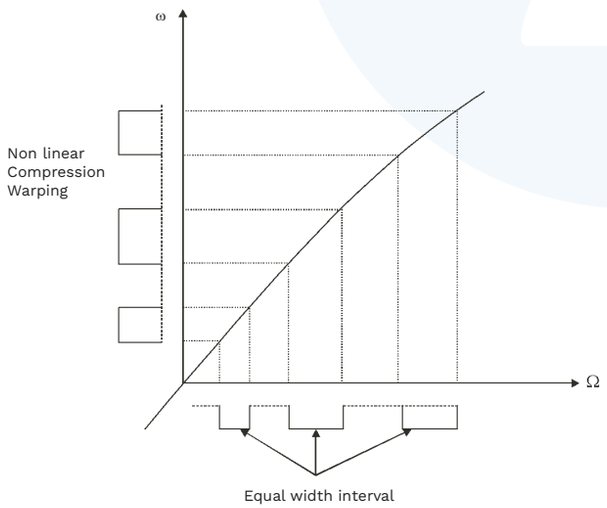


Fig. 5.28

Characteristics of Commonly Used Analog Filter

The important characteristics of commonly used analog filters and the relevant filter parameters are discussed in this section. Several frequency transformations convert a low pass prototype filter into either a band pass, high pass or band elimination filter.

Butterworth Filters

The squared magnitude function for an analog Butterworth low pass filter is:

$$|H(\Omega)|^2 = \frac{1}{1+(\Omega/\Omega_c)^{2N}} = \frac{1}{1+\epsilon^2(\Omega/\Omega_p)^{2N}}$$

$N \Rightarrow$ Order of filter

$\Omega_c \Rightarrow$ 3 dB cut-off frequency

$\Omega_p \Rightarrow$ Pass band edge frequency

$\frac{1}{1+\epsilon^2} \Rightarrow$ Band edge value of $|H(\Omega)|^2$

$$H(s)\,H(-s)\big|_{s=j\Omega} = |H(\Omega)|^2$$

$$H(s)\,H(-s) = \frac{1}{1+(-s^2/\Omega_c^2)^N}$$

The poles of H(s) H(−s) occurs on a circle of radius Ω_c at equally spaced points.

$$\frac{-s^2}{\Omega_c^2} = (-1)^{1/N} = e^{j(2k+1)\pi/N},\ k = 0.1, - - - N-1$$

$$s_k = \Omega_c e^{j\pi/2} e^{j(2k+1)\pi/2N},\ K = 0, 1, - - - N-1$$

Poles Position for N = 4

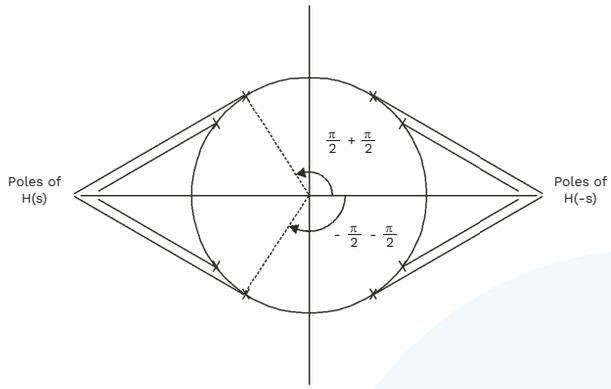


Fig. 5.29

Poles Position for N = 5

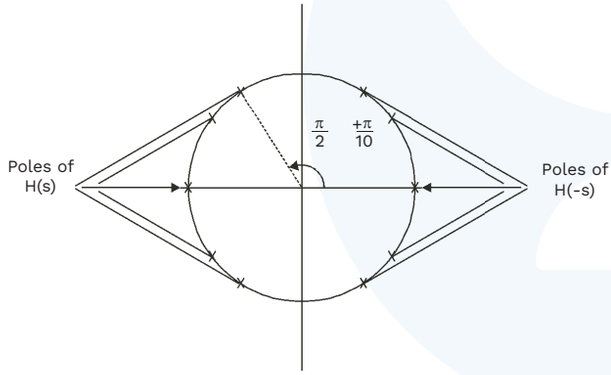


Fig. 5.30

At $\Omega = \Omega_s$

$$\frac{1}{1+\epsilon^2\left(\Omega_s/\Omega_p\right)^{2N}} = \delta_2^2$$

$$1+\epsilon^2\left(\Omega_s/\Omega_p\right)^{2N} = \frac{1}{\delta_2^2}$$

$$\epsilon^2\left(\frac{\Omega_s}{\Omega_p}\right)^{2N} = \frac{1}{\delta_2^2} - 1$$

$$\epsilon^2\left(\frac{\Omega_s}{\Omega_p}\right)^{2N} = \delta^2$$

$$\delta_2 = \frac{1}{\sqrt{1+\delta_2}}$$

$$\left(\frac{\Omega_s}{\Omega_p}\right)^{2N} = \frac{\delta^2}{\epsilon^2}$$

$$\left(\frac{\Omega_s}{\Omega_p}\right)^{2N} = \left(\frac{\delta}{\epsilon}\right)^2$$

$$2N \log\left(\frac{\Omega_s}{\Omega_p}\right) = 2 \log\left(\frac{\delta}{\epsilon}\right)$$

$$\boxed{N = \frac{\log(\delta/\epsilon)}{\log(\Omega_s/\Omega_p)}}$$

Butterworth filter is characterised by the parameters N, δ_2, ϵ and the ratio Ω_s / Ω_p.

Solved Examples

18. Determine the order and the poles of a low pass Butterworth filter, which has a –3 dB bandwidth of 500 Hz and an alternation of 40 dB at 100 Hz.

Solution:

$\Omega_c = 1000\ \pi$

$\Omega_s = 2000\ \pi$

$\alpha = 20 \log \delta_2$

$-20 \log \delta_2 = 40$

$\log \delta_2 = -2$

$\delta_2 = 10^{-2} = 0.01$

$$N = \frac{\log\left[\frac{1}{\delta_2^2} - 1\right]}{2\log\left(\Omega_s/\Omega_c\right)} = \frac{\log\left(10^4 - 1\right)}{2\log\left(\frac{2000\pi}{1000\pi}\right)}$$

$\boxed{N = 6.64}$

$N \simeq 7$

$s_k = \Omega_c e^{j\pi/2} e^{j(2k+1)\pi/2N}$

$s_k = 1000\ \pi\ e^{j\pi/2} e^{j(2k+1)\pi/14}$

$$\boxed{s_k = 1000\ \pi\ e^{j\left[\frac{\pi}{2}+(2k+1)\pi/14\right]}},\ k = 0, 1, 2, \ldots\ldots 6$$

Chebyshev Filters

There are two types of Chebyshev filter.

1. Type 1: All poles of the filter exhibit equiripple behaviour in the pass band and a monotonic characteristic in the stop band.

2. Type 2: This type of Chebyshev filter contains both poles and zeros and exhibits a monotonic behaviour in the pass band and equiripple behaviour in the stop band. The zeros lie on the $j\omega$ axis.

$$|H(\Omega)|^2 = \frac{1}{1+\epsilon^2\, T_N^2(\Omega/\Omega_p)}$$

For type 1 Chebyshev filter:

$T_N(x) \Rightarrow N^{th}$ order Chebyshev polynomial

$$T_N(x) = \begin{cases} \cos\left(N\cos^{-1}x\right) & |x| \le 1 \\ \cosh\left(N\cosh^{-1}x\right) & |x| > 1 \end{cases}$$

The Chebyshev polynomial can be generated by recursine equation.

$T_{N+1}(x) = 2xT_N(x) - T_{N-1}(x)$, N = 1, 2 ...,

Where, $T_o(x) = 1$

$T_1(x) = x$

$T_2(x) = 2xT_1(x) - T_0(x)$

$T_2(x) = 2x^2 - 1$

$T_3(x) = 4x^3 - 3x$

Properties of Chebyshev Polynomial

1. $|T_N(x)| \le 1 \; \forall \; |x| \le 1$
2. $T_N(1) = 1 \; \forall \; N$
3. All roots of the polynomial $T_N(x)$ occur in the interval $-1 \le x \le 1$.

For N odd,

$T_N(0) = 0$

$|H(0)|^2 = 1$

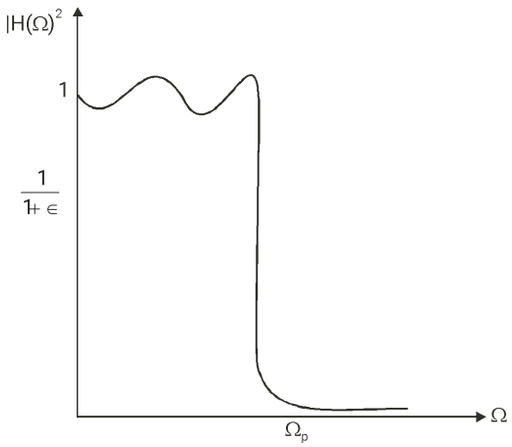


Fig. 5.31

For N Even,

$T_N(0) = 1$

$|H(0)|^2 = \frac{1}{1+\epsilon^2}$ For N odd,

$T_N(0) = 0$

$|H(0)|^2 = 1$

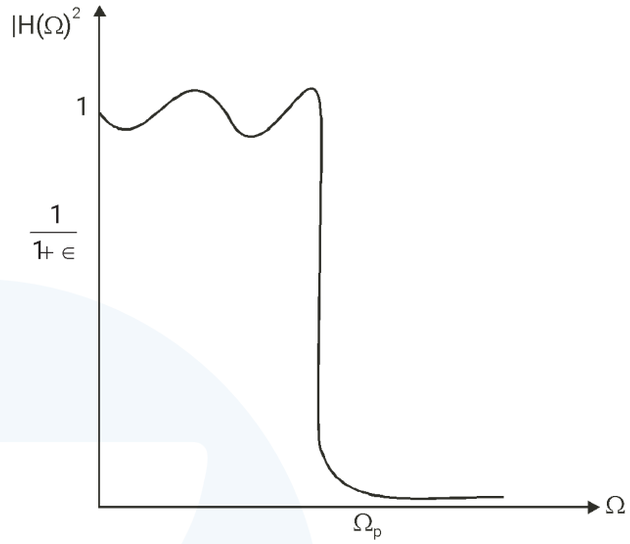


Fig. 5.32

For N-even,

$T_N(0) = 1$

$|H(0)|^2 = \frac{1}{1+\epsilon^2}$

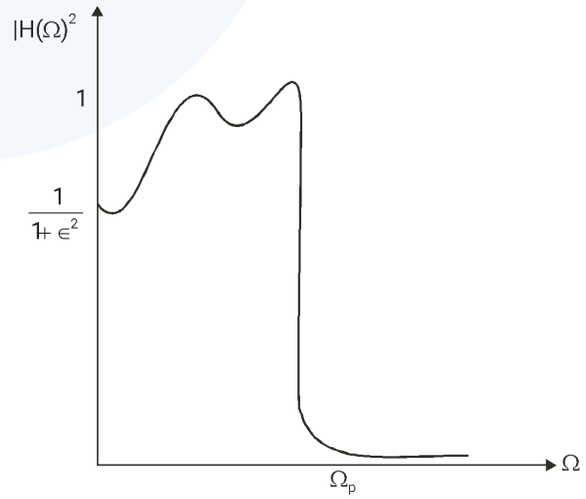


Fig. 5.33

At the band edge frequency $\Omega = \Omega_p$

$T_N(1) = 1$

$$\frac{1}{\sqrt{1+\epsilon^2}} = 1-\delta_1$$

$$\epsilon^2 = \frac{1}{(1-\delta_1)^2} - 1$$

$\delta_1 \Rightarrow$ Pass band ripple value

The poles of type 1 Chebyshev filter lie on an ellipse in the s-plane with major axis.

$$r_1 = \Omega_p\left(\frac{\beta^2+1}{2\beta}\right)$$

and minor axis,

$$r_2 = \Omega_p\left(\frac{\beta^2-1}{2\beta}\right)$$

$$\beta = \left(\frac{\sqrt{1+\epsilon^2}+1}{\epsilon}\right)^{1/N}$$

The angular position of the poles of the Butterworth filter is:

$$\phi_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{2N}, \; k = 0, 1, 2, ---N-1$$

Chebyshev filter lie on the ellipse at the co-ordination (x_k, y_k) k = 0, 1, ... N – 1

$$x_k = r_2 \cos\phi_k, \; k = 0, 1, ...N-1$$

$$y_k = r_1 \sin\phi_k, \; k = 0, 1, ... N-1$$

$$|H(\Omega)|^2 = \frac{1}{1+\epsilon^2\left[T_N^2(\Omega_s/\Omega_p)/T_N^2(\Omega_s/\Omega)\right]}$$

$\Omega_s \Rightarrow$ Stop band frequency

Zeros are located at the imaginary axis at the points.

$$s_k = j\frac{\Omega_s}{\sin\phi_k} \quad k = 0, 1, ------- N-1$$

The poles are located at the points (v_k, w_k).

$$v_k = \frac{\Omega_s x_k}{\sqrt{x_k^2+y_k^2}}, \; k = 0, 1, -----N-1$$

$$= \frac{\Omega_s y_k}{\sqrt{x_k^2+y_k^2}}, \; k = 0, 1, ---N-1$$

$$\beta = \left[\frac{1+\sqrt{1-\delta_2^2}}{\delta_2}\right]^{1/N}$$

Order of filter

$$N = \frac{\log\left[\left(\sqrt{1-\delta_2^2}+\sqrt{1-\delta_2^2(1+\epsilon^2)}\right)/\epsilon\,\delta_2\right]}{\log\left[(\Omega_s/\Omega_p)+\sqrt{(\Omega_s/\Omega_p)^2-1}\right]}$$

$$= \cosh^{-1}(\delta/\epsilon)/\cosh^{-1}(\Omega_s/\Omega_s)$$

$$\delta_2 = \frac{1}{\sqrt{1+\delta^2}}$$

Solved Examples

19. Determine the order and poles of a type 1 low pass Chebyshev filter that has a 1-dB ripple in the pass band, a cut-off frequency $\Omega_p = 1000\,\pi$, a stop band frequency of 2000 π, and an alternation of 40 dB or more for $\Omega \geq \Omega_s$.

Solution:

$10\log(1+\epsilon^2) = 1$

$1+\epsilon^2 = 10^{0.1} = 1.259$

$\boxed{\epsilon = 0.5088}$

$-20\log\delta_2 = 40$

$\delta_2 = 10^{-2} = 0.01$

$$N = \frac{\log\left[\left(\sqrt{1-\delta_2^2}+\sqrt{1-\delta_2^2(1+\epsilon^2)}\right)/\epsilon\,\delta_2\right]}{\log\left[(\Omega_s/\Omega_p)+\sqrt{(\Omega_s/\Omega_p)^2-1}\right]^{N-1}}$$

$$= \frac{\log 196.54}{\log(2+\sqrt{3})} = 4$$

$$\beta = \left[\frac{\sqrt{1+\epsilon^2}+1}{\epsilon}\right]^{1/N}$$

$$= \left[\frac{\sqrt{1+(0.5088)^2}+1}{0.5088}\right]^{1/4}$$

$\boxed{\beta = 1.429}$

$$r_1 = \Omega_p\left(\frac{\beta^2+1}{2\beta}\right) = 1.06\,\Omega_p$$

$$r_2 = \Omega_p\left(\frac{\beta^2 - 1}{2\beta}\right) = 0.365\Omega_p$$

$$\phi_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{2N}$$

$$\phi_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{8}, \; k = 0, 1, 2, 3$$

Location of the Poles

$$s_1 = x_1 + jy_1 = -0.1397\Omega_p + j0.979\,\Omega_p$$

$$s_2 = x_1 - jy_1 = -0.1397\,\Omega_p - j0.979\Omega_p$$

$$s_3 = x_2 + jy_2 = -0.337\,\Omega_p + j0.4056\,\Omega_p$$

$$s_4 = x_2 - jy_2 = -0.337\,\Omega_p - j0.4050\,\Omega_p$$

Elliptic Filters

Elliptic filter exhibits equiripple behaviour in both the pass band and the stop band.

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2\, U_N(\Omega / \Omega_p)}$$

$U_N(x)$ = Jacobian elliptic function of order N

The zeros lie on the $j\Omega$ axis.

The filter order required to achieve a given set of specifications in pass band ripple δ_1, stop band ripple δ_2 and transition ratio Ω_p / Ω_s is given as:

$$N = \frac{K(\Omega_p / \Omega_s)K\left(\sqrt{1 - (\epsilon^2 / \delta^2)}\right)}{k(\epsilon / \delta)k\left(\sqrt{1 - (\Omega_p / \Omega_s)^2}\right)}$$

$K(x) \Rightarrow$ Complete elliptic integral

$$K(x) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - x^2 \sin^2\theta}}$$

$$\delta_2 = \frac{1}{\sqrt{1 + \delta^2}}$$

Bessel Filters

Bessel filters are class of all pole filters that are characterised by the system function.

$$H(s) = \frac{1}{B_N(s)}$$

$B_N(s) \Rightarrow N^{th}$ order Bessel polynomial

$$B_N(x) = \sum_{k=0}^{N} a_k s^k$$

$$a_k = \frac{(2N-k)!}{2^{N-k}k!\,(N-k)!}, \; k = 0, 1, ----N$$

Bessel polynomial may be generated recursively from the relation

$$\boxed{B_N(s) = (2N-1)B_{N-1}(s) + s^2 B_{N-2}(s)}$$

with $B_0(s) = 0$ and $B_1(s) = s + 1$ as initial conditions.

Practice Exercise

Level I

1. Linear phase systems have a constant:
(A) phase
(B) magnitude
(C) group delay
(D) angle

2. As the length of the window increases in designing an FIR filter, the width of the main lobe:
(A) does not change
(B) increases
(C) is zero
(D) decreases

3. For a stop-band attenuation of –40 dB in a linear phase FIR filter (1,0)60, the window used is:
(A) rectangular
(B) blackman
(C) hanning
(D) hamming

4. Out of all windows available, the one which has the narrowest main lobe for a given length is:
(A) rectangular
(B) Blackman
(C) Hanning
(D) Hamming

5. In Kaiser window, the peak side-lobe is (1,0)60, but the width of main lobe is (1,0)60, respectively.
(A) variable, variable
(B) fixed, fixed
(C) fixed, variable
(D) variable, fixed

6. The deviation of group delay from a constant indicates the degree of:
(A) linearity of the phase
(B) symmetry of the phase
(C) non-linearity of the phase
(D) non-symmetry of the phase

7. Attenuation in dB is given as:
(A) $-20\log_{10} H(e^{j\omega})$
(B) – gain in dB
(C) Both (A) and (B)
(D) None of the above

8. An FIR filter is always stable because all of its pole(s) are:
(A) at the origin
(B) at the infinity
(C) at the ROC
(D) None of the above

9. The main lobe width of length M rectangular window is:
(A) $\frac{4\pi}{M}$
(B) $\frac{8\pi}{M}$
(C) $\frac{12\pi}{M}$
(D) variable

10. The main lobe width of length M Bartlett window is:
(A) $\frac{4\pi}{M}$
(B) $\frac{8\pi}{M}$
(C) $\frac{12\pi}{M}$
(D) variable

11. The main lobe width of length M Hanning windows is:
(A) $\frac{4\pi}{M}$
(B) $\frac{8\pi}{M}$
(C) $\frac{12\pi}{M}$
(D) variable

12. The main lobe width of length M Hamming window is:
(A) $\frac{4\pi}{M}$
(B) $\frac{8\pi}{M}$
(C) $\frac{12\pi}{M}$
(D) variable

13. The main lobe width of length M Blackman window is:
(A) $\frac{4\pi}{M}$
(B) $\frac{8\pi}{M}$
(C) $\frac{12\pi}{M}$
(D) variable

14. The main lobe width of length M Kaiser window is:
(A) $\frac{4\pi}{M}$
(B) $\frac{8\pi}{M}$
(C) $\frac{12\pi}{M}$
(D) variable

15. A Blackman window can eliminate ripple in FIR filers. The trade-off is:
(A) larger transition bandwidth
(B) smaller transition bandwidth
(C) a non-linear phase response
(D) possible instability

16. Coefficient symmetry is important in FIR filters because it provides:
(A) a smaller transition bandwidth
(B) less pass-band ripple
(C) less stop-band ripple
(D) a linear phase response

17. If a linear phase filter has a phase response of 40° at 200 Hz, what will be its phase response at a frequency of 400 Hz (assuming that both frequencies are in the pass-band of the filter)?
(A) 35°
(B) 45°
(C) 40°
(D) 80°

18. We would like to design an equiripple LPF of order 30. For a Type-1 filter of order 30, what is the minimum number of alterations that this filter may have and what is the maximum number?
(A) The minimum number is 17 and the maximum is 18.
(B) The minimum number is 16 and the maximum is 17.
(C) The minimum number is 30 and the maximum is 31.
(D) The minimum number is 29 and the maximum is 30.

19. What are the minimum and maximum number of external frequencies that a Type-1 BSF of order 128 must have?
(A) The minimum number is 66 and the maximum is 67.
(B) The minimum number is 66 and the maximum is 69.
(C) The minimum number is 65 and the maximum is 66.
(D) The minimum number is 65 and the maximum is 68.

20. The effect of windowing on a signals spectrum can be:
(A) reduced resolution
(B) leakage
(C) Both (A) and (B)
(D) None of the above

21. A digital filter is said to be an IIR, if:
(A) all poles are outside the unit circle
(B) it oscillates
(C) one of the denominator coefficients is non-zero
(D) the current output depends on the previous output

Level II

1. In relation to the bilinear transformation method of IIR design, which of the following applies?
(A) The warping effect gets worse as we get closer to Nyquist frequency
(B) Prewarping compensates for the distortion
(C) Prewarping compensates for the distortion
(D) Frequency scaling eliminates the effect of warping in the frequency response.

2. In the impulse-invariant design procedure, the relationship between continuous-time and discrete-time frequency is:
(A) non-linear
(B) parabolic
(C) linear
(D) exponential

3. Which filter has equiripple characteristics in the pass band and varies monotonically in the stop-band?
(A) Type-I Chebyshev filter
(B) Type-II Chebyshev filter
(C) Butterworth filter
(D) Elliptical filter

4. If a continuous time system has poles only in the left half of the s-plane, then the corresponding digital filter must have poles:
(A) only outside the unit circle
(B) only inside the unit circle
(C) anywhere on the z-plane
(D) $2 < |z| < 3$

5. The output of two digital filters can be added, or the same effect can be achieved by:
(A) adding their coefficients
(B) subtracting their coefficients
(C) convolving their coefficients
(D) averaging their coefficients

6. A digital signal processor convolves each discrete sample with four coefficients and they are all equal to 0.25. This must be:
(A) an IIR filter
(B) an RRR filter
(C) an FIR filter
(D) All of the above

7. In the impulse-invariant transformation method, the relationship between the analog frequency Ω and digital frequency ω is given by:
(A) $\omega = \Omega T$
(B) $\omega = \Omega T^2$
(C) $\Omega = \omega T$
(D) $\Omega = \omega T^2$

8. In the impulse-invariant transformation method, the relationship between analog and digital transformation is given by:
(A) $\frac{1}{s+p_k} \longleftrightarrow \frac{1}{1-e^{-p_k T}z^{-1}}$
(B) $\frac{1}{s+p_k} \longleftrightarrow \frac{1}{1+e^{-p_k T}z^{-1}}$
(C) $\frac{1}{s+p_k} \longleftrightarrow \frac{1}{1-e^{p_k T}z^{-1}}$
(D) $\frac{1}{s-p_k} \longleftrightarrow \frac{1}{1-e^{p_k T}z^{-1}}$

9. In the bilinear transformation method, the relationship between the analog frequency Ω and digital frequency ω is given by:
(A) $\Omega = \tan\frac{\omega}{2}$
(B) $\Omega = \frac{2}{T}\tan\frac{\omega}{2}$
(C) $\Omega = \frac{T}{2}\tan\frac{\omega}{2}$
(D) $\Omega = \frac{2}{T}\tan\frac{2}{\omega}$

10. In the bilinear transformation method, non-linearity in the relationship between ω and Ω is known as:
(A) frequency compensation
(B) frequency mixing
(C) aliasing
(D) frequency warping

11. The transformation method in which the impulse response of a digital filter is the sampled version of the impulse response of the analog filter is:
(A) bilinear
(B) step-invariant
(C) impulse-invariant
(D) None of the above

12. The transformation method in which a strip of width $\frac{2\pi}{T}$ in the s-plane is mapped into the entire z-plane is:
(A) bilinear
(B) impulse-invariant
(C) step-invariant
(D) None of the above

13. The mapping in the impulse-invariant method is:
(A) one-to-one
(B) one-to-many
(C) many-to-one
(D) many-to-many

14. The mapping in the bilinear transformation method is:
(A) one-to-one
(B) one-to-many
(C) many-to-one
(D) many-to-many

15. The method in which aliasing effect is not present is:
(A) bilinear
(B) step-invariant
(C) impulse-invariant
(D) matched z-transform

16. The poles of a Butterworth filter lie on a/an:
(A) circle
(B) ellipse
(C) parabola
(D) helix

17. The pole of a Chebyshev filter lie on a/an:
(A) circle
(B) ellipse
(C) parabola
(D) helix

18. The behaviour exhibited by a type 1 Chebyshev filter exhibit in the pass-band is:
(A) monotonic
(B) equiripple
(C) both (a) and (b)
(D) none of these

19. The behaviour exhibited by a type 1 Chebyshev filter in the stop-band is:
(A) monotonic
(B) equiripple
(C) Both (A) and (B)
(D) None of the above

20. The behaviour exhibited by a type 2 Chebyshev filter in the pass-band is:
(A) monotonic
(B) equiripple
(C) Both (A) and (B)
(D) None of the above

21. The behaviour exhibited by a type 2 Chebyshev filter in the stop-band is:
(A) monotonic
(B) equiripple
(C) Both (A) and (B)
(D) None of the above

22. How is the transition band of a Butterworth filter related to that of a Chebyshev filter?
(A) It is more
(B) It is equal
(C) It is less
(D) None of the above

23. For the same specifications, how is the number of poles in a Butterworth filter related to that of a Chebyshev filter?
(A) It is more.
(B) It is equal.
(C) It is less.
(D) None of the above

24. For the same specifications, how is the order of a Butterworth filter related to that of a Chebyshev filter?
(A) It is more.
(B) It is equal.
(C) It is less.
(D) None of the above

Answer Key

Level I

1. (C)	**2.** (D)	**3.** (B)	**4.** (A)	**5.** (A)	**6.** (C)	**7.** (C)	**8.** (A)	**9.** (A)	**10.** (B)
11. (B)	**12.** (B)	**13.** (C)	**14.** (D)	**15.** (A)	**16.** (D)	**17.** (D)	**18.** (A)	**19.** (B)	**20.** (C)
21. (D)									

Level II

1. (A)	**2.** (C)	**3.** (A)	**4.** (B)	**5.** (A)	**6.** (C)	**7.** (A)	**8.** (A)	**9.** (B)	**10.** (D)
11. (C)	**12.** (B)	**13.** (C)	**14.** (A)	**15.** (A)	**16.** (A)	**17.** (B)	**18.** (B)	**19.** (A)	**20.** (A)
21. (B)	**22.** (A)	**23.** (A)	**24.** (A)						